

## CORE MATHEMATICS GRADE 10: LINEAR MOTION

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.9 (8 Lessons)*

| SUB-STRAND OVERVIEW |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grade Level         | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Subject             | Core Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Strand              | Strand 2.0: Measurements and Geometry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Sub-Strand          | Sub-Strand 2.9: Linear Motion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Total Duration      | 8 lessons × 40 minutes = 320 minutes total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Sub-Strand Content  | <ul style="list-style-type: none"> <li>– Distance, displacement, speed, velocity, and acceleration — definitions, units, and real-life meaning</li> <li>– Velocity and acceleration — calculating velocity and acceleration from given data</li> <li>– Displacement-time graph — drawing displacement-time graphs from tables of values</li> <li>– Interpretation of displacement-time graph — extracting velocity, rest periods, and direction from the graph</li> <li>– Velocity-time graph — drawing velocity-time graphs from tables of values</li> <li>– Interpretation of velocity-time graph — extracting acceleration, distance travelled, and deceleration from the graph</li> <li>– Relative speed — calculating relative speed of two bodies moving in the same or opposite directions</li> <li>– Application of linear motion — solving real-life problems involving linear motion concepts</li> </ul> |
| Learning Outcomes   | <p>By the end of the sub-strand, the learner should be able to:</p> <ol style="list-style-type: none"> <li>explain the terms distance, displacement, speed, velocity and acceleration in real-life situations,</li> <li>determine velocity and acceleration in different situations,</li> <li>draw a displacement-time graph of different situations,</li> <li>interpret displacement-time graph in different situations,</li> <li>draw a velocity-time graph from given tables in different situations,</li> <li>interpret velocity-time graph in different situations,</li> <li>determine relative speed of two moving bodies in different situations,</li> <li>appreciate use of linear motion in real-life situations.</li> </ol>                                                                                                                                                                              |
| Core Competencies   | <ul style="list-style-type: none"> <li>– Communication and Collaboration: the learner discusses different displacement-time graphs with peers, presents findings from races and motion experiments, and collaborates to interpret velocity-time graphs in group activities.</li> <li>– Self-Efficacy: the learner participates in a race of a specified distance on the school field, records the time taken by each learner, and confidently works out the velocity and acceleration of different learners.</li> <li>– Critical Thinking and Problem Solving: the learner observes two bodies in motion — such as matatus when overtaking on Thika Road or Mombasa Road — and works out the relative speed of the two vehicles, reasoning about the dangers of careless overtaking.</li> </ul>                                                                                                                    |
| Core Values         | <ul style="list-style-type: none"> <li>– Responsibility: as the learner uses objects in motion such as thrown balls, rolling carts, or running peers to explain distance, displacement, speed, velocity and acceleration, and takes care of measuring equipment such as stopwatches and tape</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | <p>measures.</p> <ul style="list-style-type: none"> <li>– Integrity: as the learner participates in a race of a specified distance, records time taken by each learner honestly, and accurately works out the velocity and acceleration of different learners without falsifying results.</li> <li>– Unity: as the learner participates and collaborates in group races and data-recording exercises to record time taken by each learner and work out the velocity and acceleration of different learners.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Science &amp; Engineering Practices</b>        | <ul style="list-style-type: none"> <li>– Analysing and Interpreting Data: the learner collects time and distance data from a school running race or a rolling object, plots displacement-time and velocity-time graphs, and draws conclusions about motion patterns from the shape of the graphs.</li> <li>– Using Mathematics and Computational Thinking: the learner applies formulas for velocity (<math>v = s/t</math>) and acceleration (<math>a = \Delta v/\Delta t</math>), calculates areas under velocity-time graphs to find distance, and computes gradients of displacement-time graphs to find speed.</li> <li>– Constructing Explanations: the learner uses evidence from graphs and calculations to explain the motion of matatus, cyclists in the Tour du Rwanda, or athletes at the Nairobi Marathon, connecting mathematical results to real-world behaviour.</li> <li>– Obtaining, Evaluating, and Communicating Information: the learner presents motion data collected during the school race to the class, evaluates the accuracy of readings, and communicates findings using correctly labelled graphs and written interpretations.</li> </ul>                                                                                                                   |
| <b>Pertinent &amp; Contemporary Issues (PCIs)</b> | <ul style="list-style-type: none"> <li>– Learner Support Programmes (Sports and Games): the learner participates in a race of a specified distance on the school field, records time taken by each learner and works out the velocity and acceleration — connecting mathematics to athletics and physical education.</li> <li>– Socio-Economic and Environmental Issues (Safety and Security): the learner observes two bodies in motion such as matatus when overtaking or moving in opposite directions along Kenyan highways and works out the relative speed of these two bodies in motion, appreciating the mathematical basis of road safety regulations.</li> <li>– Road Safety: as the learner observes two vehicles when one is overtaking, discusses and works out the relative speed of the two vehicles, and also discusses the dangers of careless overtaking on Kenyan roads such as the Nairobi–Nakuru highway.</li> </ul>                                                                                                                                                                                                                                                                                                                                                |
| <b>Career Connections</b>                         | <ul style="list-style-type: none"> <li>– Traffic Engineering and Road Safety: understanding velocity, acceleration, and relative speed underpins the design of speed limits, safe overtaking zones, and traffic light timing on Kenyan roads managed by KENHA and NTSA.</li> <li>– Athletics Coaching and Sports Science: coaches at Athletics Kenya use displacement-time and velocity-time data to analyse sprint performance, pace strategy, and acceleration profiles of athletes like Faith Kipyegon and Eliud Kipchoge.</li> <li>– Aviation and Railway Transport: Kenya Airways pilots and SGR (Standard Gauge Railway) operators use principles of linear motion — including velocity, acceleration, and braking distance — to plan safe operations.</li> <li>– Physics and Engineering: a foundation in linear motion kinematics is essential for university-level physics and mechanical engineering programmes at institutions such as the University of Nairobi and Jomo Kenyatta University of Agriculture and Technology (JKUAT).</li> <li>– Logistics and Supply Chain: fleet managers at companies operating along the Northern Corridor (Mombasa–Nairobi–Kampala) use speed, distance, and time calculations to plan delivery schedules and fuel efficiency.</li> </ul> |
| <b>Focus for Lessons</b>                          | <p>This unit develops learners' ability to describe, calculate, graphically represent, and interpret linear motion using the key quantities of distance, displacement, speed, velocity, acceleration, and relative speed. Learners move from concrete experience — running a measured race on the school field and timing each other with stopwatches — through graphical modelling on displacement-time and velocity-time graphs, to abstract problem-solving involving relative speed of vehicles on Kenyan highways. Throughout the unit, Kenyan contexts such as the Nairobi Marathon, SGR train journeys, matatu overtaking on the Nakuru–Nairobi highway, and athletes at Kipchoge Keino Stadium anchor every mathematical concept in meaningful real-life experience.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Driving Question / Key Inquiry Questions</b>   | <p>DRIVING QUESTION: How can we use mathematics — calculations and graphs — to fully describe and explain the motion of any moving body, from a Nairobi Marathon runner to a matatu on the Nakuru highway?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>KICD KEY INQUIRY QUESTIONS:</p> <ol style="list-style-type: none"> <li>1. What is speed?</li> <li>2. How do we use displacement-time graphs in real life?</li> <li>3. How do we use velocity-time graphs?</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Anchoring Phenomenon</b> | <p>A VIDEO CLIP AND DATA PRINTOUT of the 2023 Nairobi Marathon is shown to the class. The clip shows elite runners — including Kenyan athletes — accelerating off the start line, settling into a steady pace along Uhuru Highway, slowing slightly at water stations, and sprinting to the finish. Alongside the clip, a printed table shows the split times (time recorded at every 5 km mark) for two runners: Runner A (who runs at a nearly constant pace) and Runner B (who starts fast, slows in the middle, and sprints at the end). Students are asked: "Both runners finished the 42 km race, but their split-time tables look completely different — Runner B was actually ahead at 30 km but Runner A won. How can we use numbers and graphs to explain exactly what happened at every point in the race?" This is puzzling because students intuitively know both runners covered the same distance, yet the patterns of their motion were completely different — and a simple calculation of total time does not reveal the story of acceleration, varying velocity, and the critical sprint finish. Students must grapple with the difference between average speed and instantaneous velocity, and with how the shape of a graph — not just its start and end points — tells the full story of motion.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Storyline Thread</b>     | <p>Lesson 1 — Anchoring Phenomenon &amp; Key Terms: Learners watch the Nairobi Marathon video clip and examine the split-time data for Runner A and Runner B. Through guided discussion, learners define and distinguish distance vs. displacement, speed vs. velocity, and are introduced to acceleration. They complete a class race on the school field (or corridor) using stopwatches and tape measures, recording their own distance-time data. The driving question is posted on the Driving Question Board (DQB).</p> <p>Lesson 2 — Calculating Velocity and Acceleration: Using data from the school race and the SGR timetable supporting phenomenon, learners practise calculating velocity (<math>v = s/t</math>) and acceleration (<math>a = \Delta v/\Delta t</math>) in different situations — uniform and non-uniform motion. Worked examples use Kenyan contexts: Kipchoge's marathon pace, a boda-boda accelerating from rest. Learners update the DQB with new understanding of "what speed actually means mathematically."</p> <p>Lesson 3 — Drawing Displacement-Time Graphs: Learners use their own school-race data and the marathon split-time table to plot displacement-time graphs on squared paper. They learn conventions (labelled axes, title, correct units) and distinguish between graphs showing constant speed (straight line), rest (horizontal line), and return journey (downward slope). Learners predict what Runner B's graph will look like before plotting it.</p> <p>Lesson 4 — Interpreting Displacement-Time Graphs: Learners interpret a set of pre-drawn displacement-time graphs representing different real-life scenarios: a matatu stopping at a stage, a runner resting, a cyclist returning home. They calculate the gradient of each segment and connect gradient to velocity. The SGR stop-data is used to interpret a multi-segment graph. Learners revise their DQB entries — now able to answer "How do we use displacement-time graphs in real life?"</p> <p>Lesson 5 — Drawing Velocity-Time Graphs: Learners are introduced to velocity-time graphs using the ball-on-a-ramp supporting phenomenon and marathon sprint-finish data. They construct velocity-time graphs from given tables of values for scenarios such as a matatu accelerating from a bus stop, cruising at constant speed, and braking. Learners identify the gradient (acceleration) and the shape of each segment.</p> <p>Lesson 6 — Interpreting Velocity-Time Graphs: Learners interpret velocity-time graphs to extract acceleration, deceleration, and total distance travelled (area under the graph — counting squares and using trapezium/rectangle area). Real-life graphs include an</p> |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <p>SGR train journey and a 400 m school race. Learners connect area under the graph back to displacement, answering why Runner B lost despite leading at 30 km. The DQB is updated to address "How do we use velocity-time graphs?"</p> <p>Lesson 7 — Relative Speed: Using the matatu-overtaking supporting phenomenon, learners explore relative speed of two bodies moving in the same direction (relative speed = difference) and in opposite directions (relative speed = sum). They solve problems involving vehicles on Kenyan highways, boats on Lake Victoria, and runners passing each other on a track. Road safety implications of high relative speed are discussed and connected to NTSA speed limits.</p> <p>Lesson 8 — Consolidation, Model Building &amp; Assessment: Learners return to the original Nairobi Marathon data and now produce a complete mathematical explanation of Runner A vs. Runner B — including calculations, a displacement-time graph, a velocity-time graph, and a written interpretation of the sprint finish. They present their models to the class, updating the DQB with final answers to all three KICD key inquiry questions. A class written assessment covers all sub-strand outcomes. Learners reflect on careers that use linear motion mathematics.</p>                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Supporting Phenomena</b> | <ul style="list-style-type: none"> <li>– A STOPWATCH DATA SHEET from a 100 m sprint on the school field: learners race in groups of four over a measured 100 m track; a timekeeper records the time at 25 m, 50 m, 75 m, and 100 m for each runner. Learners notice that the time gaps between checkpoints are not equal — the runner did not cover each 25 m segment in the same time — raising the question of how velocity changes during the race.</li> <li>– AN SGR TIMETABLE AND ROUTE MAP: the Standard Gauge Railway Madaraka Express departs Nairobi (Syokimau) at 08:00 and arrives in Mombasa at 12:30, covering approximately 480 km. Learners examine the published stop times at Emali, Kibwezi, Mito Andei, and Voi and notice the train spends different amounts of time between stations — connecting real timetable data to the concept of varying velocity and distance-time relationships.</li> <li>– A DASH-CAM VIDEO DESCRIPTION of a matatu overtaking a lorry on the Nairobi–Nakuru highway: the matatu travels at 120 km/h and the lorry at 80 km/h in the same direction. Learners discuss how fast the matatu appears to be moving relative to the lorry — introducing the concept of relative speed — and connect this to the dangers of overtaking, as regulated by NTSA Kenya.</li> <li>– A STROBE PHOTOGRAPH (or printed simulation) of a ball rolling down a ramp at Mwalimu Kariuki's physics lab: equally-spaced time intervals produce unequally-spaced position marks, showing that the ball is accelerating. Students find it puzzling that equal time steps produce unequal distances — setting up the distinction between constant speed and acceleration.</li> </ul> |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

**Licensing & Copyright:** © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.

## LESSON 1 (80 minutes): Anchoring Phenomenon & Key Terms: What Is Really Happening in the Nairobi Marathon?

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | To anchor learners in the driving question by introducing the Nairobi Marathon phenomenon, establishing the key vocabulary of linear motion (distance, displacement, speed, velocity, acceleration), and collecting first-hand motion data through a class race activity.                                                                                                                                                                                                                                                       |
| <b>Knowledge</b>            | Learners explain the terms distance, displacement, speed, velocity and acceleration in real-life situations. (KICD SLO 2.9a)                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Skills</b>               | Learners participate in a measured race, record distance-time data using stopwatches and tape measures, and calculate average speed from their own results. Learners begin to distinguish between scalar and vector quantities with examples drawn from the marathon context. (KICD SLO 2.9b — introductory; KICD Core Competency: Self-Efficacy)                                                                                                                                                                               |
| <b>Attitudes</b>            | Learners appreciate that mathematics is a precise language for describing motion, value integrity in recording accurate data, and demonstrate unity by collaborating during the class race. (KICD Values: Integrity, Unity, Responsibility)                                                                                                                                                                                                                                                                                     |
| <b>Key Inquiry Question</b> | What is speed? (KICD Key Inquiry Question 1, Sub-Strand 2.9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Purpose in Storyline</b> | This is Lesson 1 of the Linear Motion unit — the opening lesson that launches the anchoring phenomenon. Learners encounter the puzzle of Runner A vs. Runner B in the 2023 Nairobi Marathon split-time data, experience motion first-hand in a school-field race, and post the driving question on the Driving Question Board (DQB). Every subsequent lesson will return to this phenomenon to deepen understanding.                                                                                                            |
| <b>Safety Notes</b>         | During the class race on the school field or corridor: (1) Ensure the running surface is clear of stones, wet patches, and obstacles before the activity. (2) Learners with physical limitations may serve as timekeepers or data recorders — include all learners meaningfully. (3) Remind learners to warm up briefly (10 seconds of jogging on the spot) before the sprint to reduce injury risk. (4) No pushing or overtaking dangerously — this reinforces the road-safety PCI on safe overtaking discussed in the lesson. |

### B. LESSON OVERVIEW

Learners watch a short video clip of the 2023 Nairobi Marathon showing Kenyan elite runners accelerating off the start line on Uhuru Highway, settling into a steady pace, slowing at water stations, and sprinting to the finish. They examine a printed split-time table for Runner A (nearly constant pace) and Runner B (fast start, mid-race slowdown, sprint finish) and grapple with the puzzle: both covered 42 km, yet Runner A won even though Runner B was ahead at 30 km. Through guided discussion learners define and distinguish distance vs. displacement, speed vs. velocity, and are introduced to the concept of acceleration. They then conduct a class race on the school field — measured distances of 40 m and 80 m — recording their own time data with stopwatches and calculating average speed. The lesson closes by posting the driving question on the Driving Question Board and capturing learners' first questions and wonderings.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                 | Learner Experience                                                                                                                 | Teacher Moves                                                                                                                       | Sensemaking Strategy                                                                                                            | Formative Assessment Strategy                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Position vs. time graphs</a> | Learners receive the printed split-time data table for Runner A and Runner B (time recorded at every 5 km mark for the full 42 km) | <b>BEFORE CLASS:</b> Print one split-time data sheet per learner (or pair). Write the two prediction prompts on the chalkboard. Cue | Prediction Protocol — Activating Prior Knowledge: By committing predictions to paper before instruction, learners surface their | Observable indicator: Can learners identify which 5 km split shows Runner B's slowdown (km 25–35) and which shows Runner A's |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p><b>HTML:</b><br/> <a href="#">Calculating Acceleration from Velocity and Time (Flexbook)</a><br/> Source: CK-12<br/> <a href="#">Search ARES</a><br/> for similar readings</p>                                                                                                                                                            | <p>Nairobi Marathon). Working individually for 3 minutes, they study the table silently and write answers to two prompts in their exercise books: (1) 'Using only the numbers in this table, who do you think is the faster runner — and why?' (2) 'At what point in the race did the situation change between the two runners?' Learners then share predictions with a shoulder partner for 2 minutes before a whole-class share-out. No answer is marked right or wrong at this stage — all predictions are recorded on the chalkboard under the heading 'OUR PREDICTIONS'.</p> | <p>the marathon video clip to the start.   LAUNCH (0–3 min): Distribute the data sheet. Say exactly: 'Do not talk yet. Read this table quietly for three minutes. Use your pencil to circle anything that surprises you.' Set a visible timer.   INDIVIDUAL THINK TIME: Allow 3 minutes of silent reading. Circulate and scan for misconceptions — look for learners who confuse 'split time' with 'total time'.   PAIR SHARE (3–6 min): Say: 'Now turn to your neighbour. You have two minutes. Compare your circles and agree on one answer to each prompt.'   WHOLE-CLASS COLD-CALL (6–10 min): Cold-call at least 5 different learners using name cards or random selection. Ask: 'Amina, what did you predict — and what number in the table made you say that?' WAIT 12 SECONDS after each name before prompting. Record every prediction verbatim on the board — do NOT evaluate yet. After 5 responses, say: 'Hold these predictions. We are about to watch what actually happened.'</p> | <p>intuitive understanding of speed and motion. This creates cognitive dissonance when the video reveals that Runner B — who looked faster in the first half — lost the race. That dissonance is the engine that drives inquiry throughout the unit.</p>                                                                                                                                                                                                                                         | <p>consistency? Circulate during individual think time and note the exercise books of 3–4 learners who appear stuck — these learners may need scaffolding in Phase 2. Do not correct at this stage; note for follow-up.</p>                                                                                                                                                                                                                       |
| <p><b>Observe Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Intro to vectors and scalars</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> for similar videos</p> <p><b>EXERCISE:</b><br/> <a href="#">Calculating Acceleration from Velocity and Time Practice</a><br/> Source: CK-12<br/> <a href="#">Search ARES</a></p> | <p>The teacher plays the 2023 Nairobi Marathon video clip (3–5 minutes) showing: the mass start on Uhuru Highway, elite Kenyan runners including pacemakers settling into rhythm along Kenyatta Avenue, the visible slowdown at a water station near the 30 km mark, and the sprint finish. While watching, learners use a simple three-column observation table in their books: 'What I SEE happening to the runner's motion   What it MEANS about their speed   My QUESTION.' After the clip, learners spend 4 minutes completing their observation</p>                         | <p>VIDEO SCREENING (10–16 min): Dim lights if possible. Say: 'As you watch, fill in column one of your observation table — just describe the motion you see. You will fill columns two and three after.' Play the clip without pausing.   POST-VIDEO WRITING (16–20 min): Say: 'Now, in silence, complete all three columns. You have four minutes.' Circulate. Look for learners who write 'Runner B slows down' — these are ready for the vocabulary upgrade in Phase 3.   GUIDED TABLE READING (20–28 min): Project or draw the split-time table on the board. Point</p>                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Notice and Wonder with Structured Observation Table — Evidence-Based Sensemaking: The three-column table (See / Means / Question) trains learners to separate raw observation from interpretation, which is NGSS Science &amp; Engineering Practice 4 (Analysing and Interpreting Data). By labelling what the motion 'means' before they have the vocabulary, learners construct their own proto-definitions of speed and acceleration, which the teacher can then formalise in Phase 3.</p> | <p>Observable indicator: In the 'Means' column, can learners use comparative language — 'Runner B is moving faster at the start', 'Runner A barely changes speed'? Collect 5–6 books during the 4-minute writing period and photograph the observation tables for later review. Flag learners who leave the 'Means' column blank — they may need a language scaffold (sentence starter: 'This means the runner is moving _____ than before').</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>tables. The teacher then hands out (or projects) a colour-coded version of the split-time table with Runner A in blue and Runner B in red, and guides learners through reading the table row by row using a 'Notice and Wonder' protocol. Key data points highlighted: Runner B's 0–5 km split is 16 min (very fast); Runner A's 0–5 km split is 18 min (steady). At 30 km, Runner B is ahead by 4 minutes total. By 42 km, Runner A wins by 2 minutes.</p>                                                                                                                                                                                                                                                                                                          | <p>to the 30 km row. Say: 'At this exact moment in the race, Runner B is ahead. Raise your hand if your prediction said Runner B would win.' (Count hands aloud — e.g., 'I see 14 hands.') Then point to the 42 km row. Say: 'But here is the finish. Runner A won. Spend 30 seconds — look at the numbers between row 30 km and row 42 km. What changed?' WAIT 30 SECONDS. Cold-call 3 learners: 'Kipchoge, what number jumped out at you?'   TRANSITION QUESTION (28–30 min): Say: 'We can see something changed — but the table only gives us total split times. It does not tell us exactly how fast each runner was going at each moment. That is the problem mathematics helps us solve. That is why we are here.' Write on the board: 'HOW CAN NUMBERS AND GRAPHS TELL THE FULL STORY OF MOTION?'</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Calculating average velocity or speed</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Acceleration and velocity</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Using the marathon context as the reference throughout, the teacher guides learners through a whole-class concept-building discussion to define, distinguish, and record five key terms: distance, displacement, speed, velocity, and acceleration. Learners create a 'Vocabulary Anchor Chart' in their books — a two-column table with the term and a marathon-based example they write themselves. Key distinctions built: (a) Distance (scalar) = total path covered, e.g., Runner A ran 42 km of road. Displacement (vector) = straight-line change in position with direction, e.g., the Nairobi Marathon start and finish are both on Uhuru Highway, so displacement <math>\neq</math> 42 km. (b) Speed (scalar) = distance <math>\div</math> time, e.g.,</p> | <p>VOCABULARY BUILD (30–48 min): Draw a two-column table on the board: 'TERM   MARATHON EXAMPLE'. Do NOT pre-fill it. For each term: (1) Ask learners to attempt a definition from the video/table evidence. (2) WAIT 12 SECONDS. (3) Cold-call 2 learners per term. (4) Refine the definition collaboratively and write it. Say for distance vs. displacement: 'The marathon route runs from the Kenya National Theatre along Uhuru Highway and loops back. Runner A ran 42 km of road. But is the start point and the finish point the same place? Kimani, what do you think the displacement is?' WAIT 12 SECONDS. Guide to the idea that if start = finish (loop), displacement = 0. Say: 'This is why direction</p>                                                                                     | <p>Co-Constructed Vocabulary Anchor Chart — Language for Thinking: Rather than lecturing definitions, the teacher elicits proto-definitions from learners' own observations and refines them collaboratively. This is NGSS Practice 6 (Constructing Explanations). Every definition is anchored to a specific moment in the marathon video or split-time table, so vocabulary is never abstract. The first calculation task (average speed) simultaneously activates numeracy skills and reveals the limitation of averages — creating the motivation for graphs in later lessons.</p> | <p>Observable indicator: Can learners correctly distinguish the scalar/vector pairs in a quick oral check? After filling the anchor chart, pose this rapid-fire question to the whole class (hands-down, cold-call): 'Achieng — if a matatu in Nairobi travels 10 km east and then 10 km west back to the start, what is the distance? What is the displacement?' Correct answer: distance = 20 km; displacement = 0 km. If more than one-third of cold-called learners (target: call 4) confuse these, pause and use a number line drawn on the board to re-teach before the field activity.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Runner A's average speed over the race. Velocity (vector) = displacement ÷ time with direction. (c) Acceleration = change in velocity ÷ time, e.g., Runner B accelerated in the final 2 km sprint. Learners then conduct a rapid calculation: using Runner A's total time (approx. 2 hr 12 min = 132 min) and the distance 42 km, they calculate Runner A's average speed in km/min and km/h. This is the first mathematical computation of the unit.</p>                                                                                                                                                                                              | <p>matters in mathematics.'   CALCULATION TASK (48–55 min): Write on the board: 'Runner A finishes in 2 hours 12 minutes. The race is 42 km. Calculate Runner A's average speed.' Say: 'Work individually. You have 4 minutes. Show every step.' After 4 minutes, cold-call one learner to come to the board and work through the solution. Affirm correct method: <math>\text{Speed} = \text{Distance} \div \text{Time} = 42 \div 2.2 = 19.09 \text{ km/h}</math> (approx.). Say: 'This is the average. But is Runner A always moving at exactly 19 km/h? Fatuma, look at the split-time table — what does it tell you?' WAIT 12 SECONDS. Draw out the concept of varying speed leading to the need for graphs.   CONNECT TO PHENOMENON: Say: 'We now have the words. In the next lessons we will build the graphs that show the full story — not just the average, but every moment of the race.'</p> |                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Position vs. time graphs</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Velocity and Acceleration (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>After returning from the field activity (which runs concurrently with or immediately after Phase 3 — see teacher moves), the class gathers to formally post the Driving Question Board. Each learner receives two sticky notes (or torn pieces of paper). On the FIRST sticky note they write one thing they now know or understand about motion. On the SECOND sticky note they write one question they still have — something the marathon data or field race made them wonder about. Sticky notes are posted on the DQB (a dedicated section of the chalkboard or a large sheet of manila paper). The teacher reads out 5–6 questions aloud and</p> | <p>DQB SETUP (55–65 min): Before this phase, prepare the DQB space on the board. Distribute two sticky notes (or paper squares) per learner. Say: 'You have three minutes. On the YELLOW note, write one thing you now understand. On the GREEN note, write one question you still have. Be specific — use the marathon or the race we just ran as your example.' WAIT 3 MINUTES of silent writing.   POSTING (65–68 min): Invite all learners to post their notes simultaneously. Quickly scan the board. Read 5–6 questions aloud, paraphrasing where needed: 'Mutua asks: why did Runner B slow down — is that acceleration?' 'Zawadi asks: how</p>                                                                                                                                                                                                                                                  | <p>Driving Question Board — Making Thinking Visible: The DQB is a living artefact that externalises the class's evolving understanding. By separating 'what I know' from 'what I wonder', learners practise metacognition (NGSS Practice 1: Asking Questions). Clustering questions into themes previews the unit structure and gives learners agency — their own questions drive the curriculum forward.</p> | <p>Observable indicator: Are learners' 'know' sticky notes conceptually accurate (e.g., 'Speed = distance ÷ time') rather than trivially descriptive (e.g., 'Runners run fast')? Are 'wonder' sticky notes genuinely inquiry-oriented (e.g., 'How does the graph look when a runner speeds up suddenly?') rather than off-topic? Photograph the DQB at the end of class. Count: (a) number of questions related to graphs — this gauges readiness for Lessons 2–3; (b) number of questions that conflate speed and velocity — target these explicitly in Lesson 2.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>groups them into clusters: 'Questions about graphs', 'Questions about the difference between speed and velocity', 'Questions about acceleration'. The driving question is written in large letters at the top of the DQB: 'HOW CAN WE USE MATHEMATICS — CALCULATIONS AND GRAPHS — TO FULLY DESCRIBE AND EXPLAIN THE MOTION OF ANY MOVING BODY, FROM A NAIROBI MARATHON RUNNER TO A MATATU ON THE NAKURU HIGHWAY?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>do we draw a graph from the split times?' Say: 'These are exactly the questions we will answer in this unit. Every lesson, we will come back to this board and move questions from the question side to the answer side.'   DRIVING QUESTION POSTING (68–70 min): Write or unveil the pre-prepared driving question banner. Read it aloud twice — once yourself, once with the class chorally. Say: 'This is our mission for the next several lessons. Keep it in mind.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Intro to vectors and scalars</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Calculating Acceleration from Velocity and Time (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners conduct a class race on the school field (or a marked corridor) as their first evidence-gathering model-building activity. The class is organised into groups of 6–8. Each group marks out a 40 m course using a tape measure (two cones or chalk marks). Within each group: 2 learners run (one at a time), 2 learners operate stopwatches (timing at 10 m, 20 m, 30 m, 40 m intervals marked with chalk), 2 learners record data in a shared table, and remaining learners observe and note qualitative descriptions ('accelerating at the start', 'steady in the middle', 'tiring at the end'). Every learner runs at least once. Data collected per runner: time at 10 m, 20 m, 30 m, 40 m. Back in class, each learner copies their group's data table into their exercise book and calculates average speed for each 10 m interval. They label each interval as 'accelerating', 'steady', or 'decelerating' based on whether speed increased, stayed similar, or decreased. This table is explicitly labelled 'MY FIRST MOTION</p> | <p>FIELD SETUP (run during or immediately after Phase 3, approximately 35–50 min mark, adjust to school timetable): Before the lesson, mark the 40 m course on the field with chalk at 10 m intervals. Prepare a data recording template on the board (or as a printed handout): Runner Name   Time at 10 m   Time at 20 m   Time at 30 m   Time at 40 m   Average Speed (each interval).   BRIEFING (2 min): Say: 'We are going to do what the Nairobi Marathon organisers did — collect our own split-time data. Your job is to record the time at every 10 m mark, just like the marathon recorders at every 5 km. We will use this data in the next lesson to draw our first graph.' Assign roles clearly. Demonstrate stopwatch use.   RACE ACTIVITY (15 min on field): Circulate between groups. Check that timers start at the runner's first movement (not the command). Prompt data recorders: 'Write the time the moment the runner crosses the chalk line — not when you think they crossed it. Accuracy matters.' For groups with</p> | <p>First-Person Data Collection as Model Building — NGSS Practice 3 (Planning and Carrying Out Investigations) &amp; Practice 5 (Using Mathematics): By generating their own split-time data, learners experience the same data-collection challenge as marathon race organisers. Labelling each interval as 'accelerating / steady / decelerating' is a proto-model of motion — a precursor to the displacement-time and velocity-time graphs in Lessons 2–4. Explicitly titling the table 'MY FIRST MOTION MODEL' establishes the unit-long practice of iterative model revision, directly mirroring NGSS Practice 2 (Developing and Using Models).</p> | <p>Observable indicator: Can learners correctly compute speed = distance ÷ time for each 10 m interval and use the comparative values to correctly label the interval as accelerating, steady, or decelerating? Exit check — before learners leave, pose this verbal question (whole class, hands up): 'If your speed in interval 1 is 4.2 m/s and your speed in interval 2 is 5.1 m/s, are you accelerating or decelerating?' Target: at least 80% of learners respond correctly. If not, clarify before dismissal: 'Acceleration means speed is INCREASING. The number gets BIGGER. In Lesson 2 we will see this as the slope of a graph.'</p> |

|  |                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |
|--|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  | MODEL — [Date]' and will be revised and expanded as a graph in Lesson 2. | inconsistent data, ask: 'Is your average speed for interval 3 higher or lower than interval 1? What does that tell you about how your runner was moving?'   DATA ANALYSIS BACK IN CLASS (50–55 min, overlaps with Phase 3 close): Say: 'Calculate the average speed for each 10 m interval using $\text{Speed} = \text{Distance} \div \text{Time}$ . Then label each interval. Use the words from our vocabulary chart.' WAIT for individual work (4 min). Cold-call 2 learners to share their most interesting finding. Ask: 'Odhiambo, did your runner's speed change from interval 1 to interval 4? What does that remind you of in the marathon data?'   MODEL LABELLING: Say: 'Write at the top of this table: MY FIRST MOTION MODEL. In Lesson 2, we will turn this table into a graph — just like the kind of graph that could explain the marathon.' |  |  |
|--|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

#### D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes: (1) PHENOMENON ENGAGEMENT — Did the marathon split-time data create genuine puzzlement? Did learners spontaneously ask 'why' questions, or did they remain passive? If passive, consider adding a more dramatic data point (e.g., Runner B's pace at km 35 compared to km 5) to sharpen the anomaly in Lesson 1 of the next class cycle. (2) VOCABULARY GAPS — Which of the five key terms (distance, displacement, speed, velocity, acceleration) caused the most confusion? Displacement is typically the hardest for Grade 10 learners because it requires directional thinking. If more than one-third of cold-called learners confused distance and displacement, plan a 5-minute warm-up activity at the start of Lesson 2 using a number line drawn on the school field. (3) FIELD ACTIVITY LOGISTICS — Did groups collect clean, reliable data? If stopwatch errors produced obviously incorrect splits (e.g., a 10 m split faster than a world record), use these as a teaching moment in Lesson 2: 'Why might our data be unreliable? What would a scientist do?' (4) DQB QUALITY — Review the photographed sticky notes. Are learners' questions specific enough to drive the next lesson? If questions are very generic ('I wonder about speed'), prompt more specific questions at the start of Lesson 2 by modelling: 'A better question might be — what does the shape of a graph look like when Runner B is accelerating in the final 2 km?' (5) INCLUSION CHECK — Were all learners — including those with physical limitations who served as timekeepers — genuinely included in the data collection and sensemaking? Adjust roles in the next practical activity if needed.

#### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                     |                                                                                                                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What did I observe? | Watched the 2023 Nairobi Marathon video clip showing Kenyan elite runners accelerating off the start on Uhuru Highway, running at a steady pace, slowing at water stations, and sprinting to the finish. Examined the split-time table for Runner A (consistent pace) |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | and Runner B (fast start, mid-race slowdown, strong sprint) and noticed that Runner B was ahead at 30 km but Runner A won. Conducted a class race on the school field, recording our own split times at 10 m intervals using stopwatches and tape measures.                                                                                                                                                                                                                                                                                                                                           |
| <b>What did I learn?</b>                     | Distance is the total path length covered (a scalar quantity — no direction needed). Displacement is the straight-line change in position from start to finish, including direction (a vector quantity). Speed = distance ÷ time (scalar). Velocity = displacement ÷ time with direction (vector). Acceleration is the rate of change of velocity — it explains why Runner B's split times changed dramatically between 30 km and 42 km. We calculated Runner A's average speed as approximately 19 km/h, and we calculated our own average speed over each 10 m interval of the class race.          |
| <b>How does this explain the phenomenon?</b> | Runner A won the 2023 Nairobi Marathon even though Runner B was ahead at 30 km because Runner B's velocity decreased sharply in the middle kilometres while Runner A maintained a nearly constant velocity throughout. Runner B's sprint in the final 2 km was not enough to overcome the time lost. A single average speed calculation cannot reveal this — we need a graph that shows what is happening at every moment of the race. Our class race data table is our first motion model, and in the next lesson we will convert it into a displacement-time graph to see the full story of motion. |

## LESSON 2 (40 minutes): Calculating Velocity and Acceleration — Making the Numbers Tell the Story

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners move from naming the concepts of velocity and acceleration (Lesson 1) to calculating them precisely using $v = s/t$ and $a = \Delta v/\Delta t$ , applying these formulas to split-time data from the Nairobi Marathon and other Kenyan motion contexts.                                                                                                                                                                                                                             |
| <b>Knowledge</b>            | <ul style="list-style-type: none"> <li>– Define velocity as the rate of change of displacement with time (<math>v = s/t</math>) and state its SI unit (m/s or km/h)</li> <li>– Define acceleration as the rate of change of velocity with time (<math>a = \Delta v/\Delta t</math>) and state its SI unit (m/s<sup>2</sup>)</li> <li>– Distinguish between uniform motion (constant velocity, zero acceleration) and non-uniform motion (changing velocity, non-zero acceleration)</li> </ul> |
| <b>Skills</b>               | <ul style="list-style-type: none"> <li>– Calculate velocity and acceleration from given displacement, speed, and time data in different real-life situations</li> <li>– Identify and interpret intervals of uniform and non-uniform motion from a table of split-time data</li> </ul>                                                                                                                                                                                                         |
| <b>Attitudes</b>            | <ul style="list-style-type: none"> <li>– Appreciate that mathematical precision — not intuition alone — is needed to fully describe motion and explain outcomes like a marathon race result</li> <li>– Show integrity by recording and using data accurately in calculations, as modelled by Kenyan athletes and engineers</li> </ul>                                                                                                                                                         |
| <b>Key Inquiry Question</b> | What is speed, and how do we calculate velocity and acceleration mathematically?                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Purpose in Storyline</b> | Lesson 2 equips learners with the quantitative tools ( $v = s/t$ and $a = \Delta v/\Delta t$ ) needed to analyse the split-time data from the Nairobi Marathon phenomenon; learners can now calculate exactly how fast each runner was moving during each 5 km segment and how dramatically Runner B's velocity changed — building the mathematical foundation for graphing motion in Lessons 3 and 4.                                                                                        |
| <b>Safety Notes</b>         | No specific hazards. If learners conduct a brief timed walk/jog in the classroom aisle or corridor as part of the Observe phase, ensure the path is clear of bags and furniture to prevent tripping.                                                                                                                                                                                                                                                                                          |

### B. LESSON OVERVIEW

This lesson is the first dedicated calculation lesson in the Linear Motion sequence. Having established in Lesson 1 that distance, displacement, speed, velocity, and acceleration are conceptually distinct, learners now practise computing velocity and acceleration from real data. The anchoring phenomenon — the split-time tables of Runner A and Runner B in the 2023 Nairobi Marathon — provides the raw numbers. Learners calculate the average velocity for each 5 km segment for both runners and immediately see a story emerge: Runner A's velocity is nearly constant (~19 km/h throughout), while Runner B's velocity surges above 21 km/h in the first 10 km, dips to around 17.6 km/h between 25 km and 35 km, then sprints again to ~22 km/h in the final 2 km. The acceleration values between segments explain why Runner B was ahead at 30 km but lost — the late sprint was too short to overcome the time lost in the slow middle section. Additional worked examples — Eliud Kipchoge's world-record marathon pace, a boda-boda accelerating from rest on Mombasa Road, and the SGR Madaraka Express changing speed between Nairobi and Athi River — anchor the formulas in familiar Kenyan contexts.

The lesson follows the five-phase ARES framework. In the Predict phase, learners estimate Runner B's velocity during the slow middle section before calculating, activating prior knowledge of average speed from Lesson 1. In the Observe phase, learners work in pairs to apply  $v = s/t$  to each 5 km split for both runners, generating a velocity table. In the Explain phase, the teacher introduces acceleration formally as  $a = \Delta v/\Delta t$  and learners calculate the acceleration between selected segments. The DQB phase captures the new insight that a single average speed hides the internal structure of motion, and that acceleration — not just velocity — determines who wins. The Model Building phase has learners annotate their personal motion model (begun in Lesson 1) with calculated values, distinguishing uniform from non-uniform sections.

Throughout, the teacher uses targeted cold-calling, mini-whiteboards (or exercise books held up), and pair-share routines to check procedural accuracy and conceptual understanding simultaneously. The lesson closes with learners writing one sentence on their

DQB sticky note answering: "What does speed actually mean mathematically?" — directly advancing the driving question.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Calculating average velocity or speed</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Setting up problems with constant acceleration</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners are shown the split-time table from Lesson 1 (projected or on a printed handout) and asked to estimate — without calculating — which 5 km segment they think Runner B ran slowest, and what velocity they guess (in km/h) for that segment. Each learner writes their estimate in their exercise book and shares briefly with their shoulder partner. The class hears three or four estimates read aloud before any calculation begins.                                                               | Display the split-time table and say: 'Before we touch a calculator or do any arithmetic, look at Runner B's times between 25 km and 35 km. In your book, write: (a) which segment you think is the slowest, and (b) your best guess for the velocity in km/h for that segment. You have 60 seconds — work alone first.' [WAIT 60 seconds, circulate, glance at books.] After 60 seconds: 'Turn to your neighbour and compare your guesses. Do you agree? 30 seconds.' [WAIT 30 seconds.] Cold-call 3 learners — choose one high estimate, one low estimate, and one middle estimate — and write the three guesses on the board as a range. Say: 'By the end of today's lesson, we will know the exact answer. Keep your guess written down — we will come back to it.' | Anchored Prediction (Estimate-Before-Calculate): Learners commit a numerical estimate to paper before instruction, creating cognitive dissonance if their calculation later differs. This makes the Observe phase calculation feel purposeful — learners are testing their own prediction — and builds number-sense for reasonable velocity magnitudes.                                                                                                                | Teacher scans exercise books during the 60-second individual think time, noting whether learners are attempting a rough division (distance ÷ time) or guessing arbitrarily. If most learners write answers between 10 and 30 km/h without any working, this confirms they have some intuitive feel for marathon speeds from Lesson 1. If estimates are wildly outside this range (e.g., 200 km/h or 2 km/h), the teacher notes these learners for targeted support during the Observe phase. |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Acceleration</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Setting up problems with constant acceleration</a><br><i>Related topic: not</i>                                                                                                                                                                                                                                            | Working in pairs, learners receive (or refer to) the printed split-time table for Runner A and Runner B. Using the formula $v = s/t$ (velocity = displacement ÷ time for each 5 km segment), each pair completes a structured velocity table for all segments of both runners. They record each velocity in km/h, rounded to one decimal place. Pairs then circle the three fastest and three slowest segments for Runner B and compare their completed velocity table with the prediction made in the Predict | Distribute or direct learners to the split-time data handout. Say: 'We are going to use the formula $v = s/t$ to find the velocity for every 5 km segment. Remember — for each segment, the displacement $s$ is 5 km and the time $t$ is the time Runner A or Runner B took to cover that specific 5 km, not the total race time. Write the formula at the top of your table before you begin.' [WAIT 15 seconds for learners to write the formula.] 'Work with your partner — one person calculates Runner A, the                                                                                                                                                                                                                                                      | Data Table Analysis with Phenomenon Connection: Learners generate their own velocity table from raw split-time data rather than receiving pre-calculated values. This mirrors the practice of NGSS Science and Engineering Practice 4 (Analysing and Interpreting Data). Comparing the velocity table back to the prediction activates metacognition. Labelling segments as uniform or non-uniform bridges the calculation to the conceptual vocabulary from Lesson 1. | Teacher checks three specific indicators during circulation: (1) Are learners dividing by the segment time (correct) or the cumulative time (common error)? (2) Are units correct — km/h rather than km/min? (3) Does the completed velocity table show Runner B with a dip between 25–35 km and a spike in the final 2 km? A correct velocity table for Runner B should show approximately: Seg 1–5 km: ~21 km/h, Seg 5–10 km: ~21.5 km/h, Seg 10–15 km: ~20 km/h, Seg 15–                  |

|                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                 | <p>phase.</p>                                                                                                                                                                                                                                                                                                                                              | <p>other calculates Runner B, then compare your tables. You have 8 minutes.' [WAIT 8 minutes, circulate actively.] After 4 minutes, pause and do a mid-task check: cold-call 1 pair and ask: 'What velocity did you get for Runner B between 5 km and 10 km?' If the answer is incorrect, ask: 'What did you use as your time for that segment — was it the cumulative time or the split time for just that 5 km?' [WAIT 10 seconds.] Redirect as needed. At 8 minutes: 'Now compare your velocity tables with the guess you made at the start. How close were you? Put a tick if your guess was within 2 km/h, a cross if it was further away.' Cold-call 2 pairs to share their comparison. Then say: 'Look at the velocity column for Runner A. What pattern do you notice?' [WAIT 15 seconds.] Cold-call 3 learners. Expected responses: 'The velocities are nearly the same each segment' / 'Runner A is consistent.' Write on the board: UNIFORM MOTION — velocity stays (approximately) constant. Then: 'Now look at Runner B. What do you see?' [WAIT 10 seconds.] Cold-call 3 different learners. Expected: 'The velocity goes up and down a lot.' Write: NON-UNIFORM MOTION — velocity changes.</p> |                                                                                                                                                                                                                                                                                                                          | <p>20 km: ~20 km/h, Seg 20–25 km: ~20 km/h, Seg 25–30 km: ~17.6 km/h, Seg 30–35 km: ~17.6 km/h, Seg 35–40 km: ~19 km/h, Seg 40–42 km: ~22 km/h. Any pair whose table diverges significantly is flagged for Explain-phase support.</p>                                                                                          |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b> <a href="#">Average velocity for constant acceleration</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> | <p>The teacher formally introduces acceleration using <math>a = \Delta v / \Delta t</math>, connecting it to the velocity changes learners just observed in Runner B's table. Learners follow two teacher-led worked examples (Kipchoge's marathon pace and a boda-boda on Mombasa Road), then independently solve two practice problems — one finding</p> | <p>Begin by pointing to the velocity table on the board and say: 'Runner B's velocity changed dramatically between segments. We have a name and a formula for how quickly velocity changes — it is called ACCELERATION.' Write on board: <math>a = \Delta v \div \Delta t</math>, where <math>\Delta v</math> = change in velocity (<math>v_{\text{final}} - v_{\text{initial}}</math>) and <math>\Delta t</math> = time taken for</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Worked Example → Guided Practice → Independent Practice (I Do – We Do – You Do): The teacher models with full narration (Kipchoge, boda-boda), then learners apply independently to phenomenon data (Runner B) and a new Kenyan context (SGR). This gradual release embeds the formulas in multiple real contexts</p> | <p>During independent practice, teacher collects three observable indicators: (1) Correct formula written before substitution — if absent, learner has not internalised the structure. (2) Unit conversion attempted for km/h to m/s or km/h/min — a common sticking point; note learners who skip this step. (3) Physical</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                    |                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p> <b>EXERCISE:</b><br/> <a href="#">Setting up problems with constant acceleration</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>acceleration for Runner B between the slow and sprint sections, and one involving the SGR Madaraka Express. Learners record worked examples and practice solutions in their exercise books.</p> | <p>that change. Say: 'The unit of acceleration is metres per second squared, <math>\text{m/s}^2</math>, or kilometres per hour per second, <math>\text{km/h/s}</math>, depending on the units we use. Let us do a worked example.'</p> <p><b>WORKED EXAMPLE 1 —</b><br/> Kipchoge's Finish Sprint: 'Eliud Kipchoge, the greatest marathon runner in history, increased his velocity from 20.0 <math>\text{km/h}</math> to 22.5 <math>\text{km/h}</math> in the final 600 m of a race, taking approximately 96 seconds. What was his acceleration?' Write on board step by step: <math>\Delta v = 22.5 - 20.0 = 2.5 \text{ km/h}</math>. Convert: <math>2.5 \text{ km/h} = 2500 \text{ m} / 3600 \text{ s} \approx 0.694 \text{ m/s}</math>. <math>\Delta t = 96 \text{ s}</math>. <math>a = 0.694 \div 96 \approx 0.0072 \text{ m/s}^2</math>. Say: 'That seems small — but for a runner after 42 km, any acceleration is extraordinary. This is why Kipchoge is remarkable.'<br/> [Pause 10 seconds — allow learners to finish copying.]</p> <p><b>WORKED EXAMPLE 2 —</b> Boda-boda on Mombasa Road: 'A boda-boda accelerates from rest at a stage near the Nyayo Stadium roundabout, reaching a velocity of 36 <math>\text{km/h}</math> in 10 seconds. Calculate its acceleration in <math>\text{m/s}^2</math>.' Write: <math>v_{\text{initial}} = 0 \text{ m/s}</math> (rest). <math>v_{\text{final}} = 36 \text{ km/h} = 36000/3600 = 10 \text{ m/s}</math>. <math>\Delta t = 10 \text{ s}</math>. <math>a = (10 - 0) \div 10 = 1 \text{ m/s}^2</math>. Say: 'That is a much larger acceleration — the boda-boda changes velocity much faster than Kipchoge did.' Cold-call 1 learner: 'Tell me in words what acceleration of <math>1 \text{ m/s}^2</math> means.' [WAIT 15 seconds.] Expected: 'Every second, the velocity increases by 1 <math>\text{m/s}</math>.' Affirm and write this on the board as a sentence definition.</p> | <p>before learners are assessed, strengthening transfer. NGSS Practice 5 (Using Mathematics and Computational Thinking) is explicitly enacted.</p> | <p>interpretation written in words — learners who can only produce a number but not interpret it (e.g., 'the velocity increased') have surface-level procedural skill without conceptual understanding. Teacher targets these learners during the DQB phase with a probing question.</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                  |                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                 |                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                  |                                                                                                                                                                                                          | <p>PRACTICE PROBLEMS — say: 'Now you will try two problems on your own. Write the formula, substitute the values, show all working, state the unit.' [WAIT 10 seconds for learners to prepare.] Problem 1: 'Using Runner B's velocity table, calculate the acceleration between the 30–35 km segment (<math>v \approx 17.6</math> km/h) and the 40–42 km segment (<math>v \approx 22</math> km/h). The time for this change was approximately 48 minutes. Is this positive or negative acceleration? What does it mean physically?' Problem 2: 'The SGR Madaraka Express leaves Nairobi Station at rest and reaches a cruising velocity of 120 km/h in 4 minutes. Calculate the acceleration in km/h/min and describe the motion.' [WAIT 6 minutes for individual work. Circulate — note errors in <math>\Delta v</math> sign convention and unit conversion.] After 6 minutes, cold-call 2 learners to present their solutions on the board. Peer-correct using the protocol: 'Formula correct? Substitution correct? Unit stated? Physical interpretation given?' Close the Explain phase by saying: 'We now have two mathematical tools: <math>v = s/t</math> and <math>a = \Delta v/\Delta t</math>. Together, these tell the complete mathematical story of any moving object — including why Runner B lost despite being ahead at 30 km.'</p> |                                                                                                                                                                                 |                                                                                                                                                                                         |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Calculating</a> | Each learner writes one sticky note (or a folded piece of paper if sticky notes are unavailable) answering the prompt: 'What does speed actually mean mathematically, and what NEW question does today's | Say: 'Look at our Driving Question: How can we use mathematics — calculations and graphs — to fully describe and explain the motion of any moving body? Today we have added two powerful tools: $v = s/t$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Driving Question Board — Progressive Claim Building: The DQB functions as a public, cumulative record of the class's growing understanding. Posting under three headers (Know / | Teacher reads all DQB notes after class. Key indicators: (a) Does the learner's 'KNOW' statement correctly use the word velocity (vector) rather than just speed (scalar)? (b) Does the |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">average velocity or speed</a></p> <p>Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Setting up problems with constant acceleration</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>calculation raise for you about the marathon runners?' Learners post their notes on the class DQB under one of three column headers: WHAT WE NOW KNOW   WHAT WE ARE STILL WONDERING   WHAT WE NEED TO FIND OUT. The class briefly reads several contributions aloud.</p>                                                                                                                                                                                                                                                                                                                                | <p>and <math>a = \Delta v / \Delta t</math>. On your sticky note, write your answer to this prompt —' [Write on board]: 'Speed means mathematically... AND I am now wondering...'. [WAIT 2 minutes for writing.] Collect/post notes. Read aloud 4–5 notes — deliberately choose: one that correctly defines velocity, one that expresses wonder about graphs ('but how do we show this visually?'), one that raises a question about negative acceleration ('what if velocity decreases — is acceleration negative?'), and one that connects to the phenomenon ('I want to know at exactly what point Runner B started slowing down'). After reading, say: 'These questions on the right column — about graphs and about exact turning points — are exactly what Lessons 3 and 4 will answer. Leave them on the board. They are our roadmap.' Circle or highlight DQB notes that mention graphs, turning points, or the word 'deceleration' — these will anchor Lesson 3's launch.</p> | <p>Wondering / Need to Find Out) externalises metacognition and makes the sequence of the unit visible to learners. Teacher curation of selected notes models scientific discourse — not all ideas are equally specific or useful, and the teacher explicitly names why certain questions are productive for the investigation.</p>                                                                                                                                                   | <p>'WONDERING' question point forward to graphs, instantaneous velocity, or deceleration — showing they are thinking beyond today? (c) Are any notes blank or trivially vague ('I wonder about speed')? These learners need a brief check-in at the start of Lesson 3. The teacher photographs the DQB for the ARES platform record.</p>                                                                                                                                                                                                                                            |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Acceleration</a></p> <p>Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Setting up problems with constant acceleration</a></p> <p><i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan</p>                                            | <p>Learners return to the personal motion model they began in Lesson 1 — a labelled diagram or annotated timeline of the Nairobi Marathon phenomenon. They now add a second layer to the model: above or beside the timeline, they write the calculated velocity for each 5 km segment of Runner B, and annotate at least two points with the acceleration value and a label (e.g., 'DECELERATING here — a is negative' or 'ACCELERATING here — a is positive'). Learners also add a key to their model distinguishing UNIFORM MOTION sections (Runner A, and Runner B's first 10 km) from NON-UNIFORM</p> | <p>Say: 'Open your Lesson 1 model — your timeline or diagram of the marathon. You built it with words and arrows. Today we are going to upgrade it with numbers.' Display a partially completed model on the board (teacher-drawn, deliberately incomplete — missing acceleration annotations). Say: 'I have added the velocity numbers above the timeline. What is missing from my model that would help someone understand Runner B's story better?' [WAIT 15 seconds.] Cold-call 2 learners. Expected: 'The acceleration values' / 'Labels for where the runner is speeding up or slowing down.' Say: 'Exactly. Add those two things to your own</p>                                                                                                                                                                                                                                                                                                                                | <p>Cumulative Model Revision (Progressive Annotation): Learners do not build a new model each lesson — they return to and deepen the same model, adding new conceptual and quantitative layers. This mirrors how scientists revise explanatory models as evidence accumulates (NGSS Practice 2: Developing and Using Models). The teacher's deliberately incomplete board model prompts learners to identify what is missing, rather than simply copying — a higher-order demand.</p> | <p>Teacher looks for three features in each updated model: (1) At least 6 velocity values correctly placed on the Runner B timeline — confirms calculation accuracy from the Observe phase. (2) At least one correctly labelled positive acceleration and one negative acceleration interval — confirms understanding that acceleration has direction/sign. (3) A key or legend that distinguishes uniform from non-uniform motion — confirms conceptual integration. Models missing any of these three features are flagged for the teacher's lesson-start review in Lesson 3.</p> |

|                                                                                                  |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |  |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <p>Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>MOTION sections (Runner B, 25–35 km and 40–42 km).</p> | <p>model now. Also add a key — a small box in the corner that says UNIFORM MOTION (constant velocity) and NON-UNIFORM MOTION (changing velocity), and shows which sections of the race each runner was in. You have 5 minutes.' [WAIT 5 minutes. Circulate and stamp/initial completed models as evidence of engagement.] With 1 minute remaining, say: 'Share your updated model with your partner. Check: does their model show at least one negative acceleration (deceleration) and one positive acceleration? If not, help them add it.' Close by saying: 'Your model is now a mathematical document — it does not just show what happened, it shows how fast and how quickly the velocity changed. Every time we learn something new in this unit, we will come back and add another layer.'</p> |  |  |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

## D. TEACHER REFLECTION

1. During the Observe phase, how many pairs were dividing by the cumulative race time rather than the segment time? What misconception about 'average' does this reveal, and how will I address it explicitly at the start of Lesson 3 before introducing displacement-time graphs?

2. When learners calculated acceleration for Runner B between the slow section and the sprint finish, did they correctly treat the change in velocity as positive (speeding up)? How many learners wrote a negative value, and what does this tell me about their understanding of direction and sign convention in vector quantities?

3. Were the Kenyan contexts (Kipchoge, boda-boda, SGR) genuinely motivating for learners, or did some learners struggle to connect them to the marathon phenomenon? Should I add a third worked example that stays entirely within the marathon data to strengthen the phenomenon thread?

4. Looking at the DQB sticky notes: how many learners spontaneously used the word 'graph' or 'plot' in their WONDERING column? This will tell me how much scaffolding is needed to motivate Lesson 3's displacement-time graph construction.

5. Were learners able to give a verbal/physical interpretation of their acceleration values (e.g., 'the runner's velocity increased by 0.7 km/h every minute'), or did they produce only numerical answers? How will I build the habit of physical interpretation into Lessons 3 and 4 so that graph gradients are understood as rates, not just slopes?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>What did I observe?</b></p> | <p>Learners generated a velocity table for Runner A and Runner B by applying <math>v = s/t</math> to each 5 km split of the 2023 Nairobi Marathon data. They observed that Runner A maintained a nearly constant velocity (~19 km/h) throughout — uniform motion — while Runner B's velocity varied from ~17.6 km/h in the 25–35 km section to ~22 km/h in the final sprint — non-uniform motion. They also calculated the acceleration of a boda-boda on Mombasa Road (<math>1 \text{ m/s}^2</math>) and the SGR Madaraka Express, and used <math>a = \Delta v/\Delta t</math> to</p> |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | find Runner B's acceleration between the slow and fast sections of the race.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>What did I learn?</b>                     | Velocity is calculated as $v = s/t$ (displacement divided by time) and is measured in m/s or km/h. Acceleration is calculated as $a = \Delta v/\Delta t$ (change in velocity divided by time taken) and is measured in $m/s^2$ . Uniform motion means constant velocity and zero acceleration; non-uniform motion means changing velocity and non-zero acceleration. A positive acceleration means the object is speeding up; a negative acceleration (deceleration) means it is slowing down. Runner A's near-constant velocity explains his consistent split times; Runner B's large positive and negative accelerations explain why his split times varied so dramatically.                                                            |
| <b>How does this explain the phenomenon?</b> | The split-time data from the Nairobi Marathon phenomenon can now be expressed as a sequence of velocities and accelerations — not just a single average speed. Runner B was ahead at 30 km because his early high velocity (positive acceleration off the start) built a time advantage. However, his large negative acceleration (deceleration) between 25 km and 35 km cost him more time than his final sprint could recover. Runner A's near-zero acceleration — uniform motion — meant every segment was equally efficient, and this consistency ultimately won the race. The mathematical tools $v = s/t$ and $a = \Delta v/\Delta t$ reveal the internal structure of motion that a single total-time comparison completely hides. |

## LESSON 3 (80 minutes): Drawing Displacement-Time Graphs: Reading the Shape of the Race

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners plot and interpret displacement-time graphs using the Nairobi Marathon split-time data and their own school-race data, connecting graph shape to the physical motion of the runners.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Knowledge</b>            | By the end of the lesson, learners will know: (a) the conventions for drawing a displacement-time graph (labelled axes, correct units, appropriate scale, title); (b) that a straight line on a displacement-time graph represents constant speed; (c) that a horizontal line represents rest (zero velocity); (d) that a steeper line represents higher speed; and (e) that a downward slope represents motion back towards the starting point.                                                                                                                                                                                                                                                                                                            |
| <b>Skills</b>               | By the end of the lesson, learners will be able to: (a) plot a displacement-time graph accurately on squared paper from a given data table; (b) predict the general shape of a displacement-time graph before plotting; (c) interpret sections of a displacement-time graph to describe the motion at each stage; and (d) compare two displacement-time graphs plotted on the same axes to explain the outcome of the Nairobi Marathon race.                                                                                                                                                                                                                                                                                                                |
| <b>Attitudes</b>            | Learners will: (a) appreciate that a graph communicates the full story of motion far more richly than a single number; (b) demonstrate integrity by plotting points accurately and not 'tidying up' a curved section into a straight line; and (c) demonstrate unity by collaborating in pairs during the plotting activity and respecting each other's observations.                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Key Inquiry Question</b> | How do we use displacement-time graphs in real life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Purpose in Storyline</b> | In Lesson 1 learners measured and compared distance, displacement, speed, and velocity using their own school-race data. In Lesson 2 they calculated average speed and velocity for Runner A and Runner B from the marathon split-time table. In this lesson (Lesson 3) they move from numerical calculations to graphical representation: they plot both runners' split-time data on the same displacement-time graph, predict the shape before plotting, and use the graph's shape — not just its endpoints — to explain why Runner A won despite Runner B leading at 30 km. This is the first time learners see that the gradient of a line segment equals speed, preparing Lesson 4's formal treatment of gradient and Lesson 5's velocity-time graphs. |
| <b>Safety Notes</b>         | No hazardous materials. Ensure learners use rulers with clean edges to avoid inaccurate plotting. Remind learners not to press so hard with pencils that they tear squared paper. If compasses are used to mark arcs, remind learners to handle pointed ends carefully.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### B. LESSON OVERVIEW

Learners open with a sketch-prediction task: without any calculation, they sketch what they think Runner A's and Runner B's displacement-time graphs will look like, annotating with reasons. They then gather evidence by plotting both runners' split-time tables (from the Nairobi Marathon data printout introduced in Lesson 1) on the same pair of axes on squared paper, following taught conventions. A structured class discussion unpacks what different line shapes (steep, gentle, horizontal, downward) mean physically. Learners return to their predictions, annotate discrepancies, and use the completed graph to write a 'race commentary' explaining exactly when and why Runner B lost the lead. The lesson closes with a Driving Question Board update and a one-sentence exit ticket linking graph gradient to speed.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                               | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Position vs. time graphs</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>HTML:</b><br><a href="#">Position-Time Graphs (Flexbook)</a><br>Source: CK-12<br> <a href="#">Search ARES</a> for similar readings         | Each learner receives a blank displacement-time axis template (displacement in km on the y-axis, time in minutes on the x-axis, both unlabelled deliberately). Working individually for 5 minutes, learners sketch — without plotting exact points — what they expect Runner A's and Runner B's lines to look like, based only on what they already know: Runner A runs at a nearly constant pace; Runner B starts fast, slows in the middle, and sprints at the end. Learners annotate each sketch with at least two reasons justifying the shape they drew. They fold and keep their prediction sketches to compare after plotting.                                                                                                  | Distribute the blank axis templates. Say: 'Before we touch the data table, I want you to be a mathematician — make a prediction. You already know from Lesson 2 how fast each runner was at different stages. Now show me what that LOOKS like as a picture. Work alone for 5 minutes.' Circulate and observe without correcting. After 5 minutes, cold-call 3 learners: 'Achieng, hold up your sketch and tell us ONE thing you drew for Runner B.' Apply 10 seconds wait time before calling. Repeat for 2 more learners. Do NOT reveal answers. Say: 'We are going to find out if your predictions are right. Keep your sketches — you will need them at the end of the lesson.' Write on the board: PREDICTION SEALED. Do not discuss correctness yet. | Predict-first sketching (pre-assessment elicitation). By committing a sketch to paper before instruction, learners activate and expose their existing mental models of motion. This creates cognitive dissonance when the plotted graph differs from their expectation, which is the engine of sensemaking. The teacher's deliberate silence about correctness sustains productive uncertainty.                                                    | Observable indicator: every learner has produced a sketch with at least one labelled annotation before Phase 2 begins. Teacher scans for two common misconceptions to address later: (1) learners who draw Runner B's line as always above Runner A's (confusing 'ahead in distance' with 'faster'); (2) learners who draw a curve where they should draw connected straight line segments. Note names of learners holding each misconception for targeted cold-calling in Phase 3.                                                                                                                                        |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Position vs. time graphs</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Position-Time Graphs Practice</a><br>Source: CK-12<br> <a href="#">Search ARES</a> for similar readings | Learners work in pairs. Each pair receives: (a) the Nairobi Marathon split-time data printout (first seen in Lesson 1) showing Runner A and Runner B's cumulative displacement (km) at every 5 km checkpoint and the corresponding elapsed time (minutes); (b) a sheet of A4 squared paper; (c) a ruler and two coloured pencils (e.g. blue for Runner A, red for Runner B). Step 1 (8 min): The teacher models — on the board using a large grid — how to set up the axes: x-axis = Time (minutes), scale 0–140 min, 1 cm = 10 min; y-axis = Displacement from start (km), scale 0–42 km, 1 cm = 5 km. Learners copy axes, add a title ('Displacement-Time Graph: 2023 Nairobi Marathon — Runner A vs Runner B'), and label axes with | At the board, demonstrate axis setup explicitly: 'Watch — I am choosing my scale so ALL my data fits AND I can read the graph clearly. Time goes on the horizontal axis — always. Displacement goes on the vertical axis — always. My title tells the reader exactly what this graph shows.' Plot the first two points for Runner A on the board while narrating each step. Then say: 'Now YOUR turn — plot all of Runner A's points first, connect them, then do Runner B. You have 20 minutes.' Circulate every 90 seconds. At 8 minutes, pause the class: 'Hold up your paper — I want to see Runner A's line. Good. Now continue with Runner B.' If a pair has a wildly incorrect scale, crouch beside them and ask: 'Your                             | Dual-representation plotting (concrete-to-graphical bridging). Learners simultaneously hold the numerical data table and the emerging graph, building the connection between a row of numbers and a point on a plane. Using two colours for the two runners on the same axes allows direct visual comparison — a strategy that makes the 'Runner B was ahead at 30 km' phenomenon visible as a spatial relationship rather than an abstract claim. | Observable indicators: (1) Both lines are plotted with correct units and a title — teacher checks during circulation. (2) The crossing point is correctly identified between the 30 km and 35 km checkpoints (roughly $t = 95\text{--}105$ min, $d = 31\text{--}33$ km). (3) Line segments are straight between checkpoints (not smoothed into a curve) — if learners draw a curve, ask: 'Do we KNOW what happened between the 30 km and 35 km checkpoints? We only have data at those two points — so we connect them with a straight line.' Record pairs who struggle with scale choice for targeted support in Phase 3. |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>units. Step 2 (12 min): Pairs plot all data points for Runner A (blue) and connect with straight line segments between consecutive checkpoints. Step 3 (8 min): Pairs plot Runner B's points (red) on the SAME axes and connect. Step 4 (2 min): Learners circle in pencil the point on the graph where Runner A's line crosses Runner B's line (the overtake during the 30–35 km stretch of the slowdown). This is the graphical answer to the driving question phenomenon.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>highest y-value is 42 km — how many squares do you have? What scale will let everything fit?' (Do NOT draw it for them.) At 18 minutes, cold-call 2 pairs to describe in one sentence what they notice about where the two lines cross. Apply 12 seconds wait time. Say: 'Circle that crossing point — that is the moment we need to explain.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Comparing linear rates example</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Position-Time Graphs (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Whole-class discussion and individual written analysis. Learners use their completed dual graph to answer five structured questions written on the board (or printed on an analysis strip): Q1 — Which section of Runner A's graph is steepest? What does that tell you about their speed at that point? Q2 — Find a section of Runner B's graph that is less steep than the section before it. What was Runner B doing? Q3 — If Runner B had completely stopped at a water station, what would their graph look like at that point? Q4 — Using your graph, estimate the time (in minutes) at which Runner B was furthest ahead of Runner A. Q5 — Write a 'race commentary' of three sentences describing Runner B's race using only the language of graph shape (steep, gentle, horizontal). Learners write answers individually (8 min), then share with their pair (3 min). Teacher leads a whole-class debrief (9 min). Learners then retrieve their original prediction sketches, compare with actual graph, and annotate in a third colour: 'I predicted correctly</p> | <p>Post the five questions. Say: 'These questions can ALL be answered by looking at your graph — you do not need to calculate anything yet. Read the graph like you read a story.' Give 8 minutes individual work. Circulate and identify at least one learner who has correctly linked steepness to speed (to affirm publicly) and one who has answered Q3 by drawing a horizontal line in the air (to cold-call). After 8 minutes: 'Share your Q1 answer with your partner. 30 seconds.' Then cold-call: 'Kipchoge — for Q1, which section did you find steepest for Runner A? Tell us why that means high speed.' [Wait 12 seconds.] Affirm: 'Exactly — steeper slope, same time interval, MORE displacement covered. That IS speed.' For Q3, ask: 'Wanjiru, if Runner B stopped completely, what does the graph look like at that moment?' [Wait 10 seconds.] Draw a small horizontal segment on the board: 'A horizontal line means displacement is NOT changing — the runner is stationary.' Then: 'Get out your original prediction sketches. Annotate them right now</p> | <p>Graph-as-narrative strategy (slope = story). Learners are explicitly taught to 'read' a graph as a sequence of events rather than as a shape to be memorised. By writing a three-sentence race commentary using graph-shape vocabulary, they verbalise the connection between gradient magnitude and speed — the key conceptual bridge to Lesson 4's formal gradient calculation. The prediction-comparison annotation makes metacognitive revision visible and valued.</p> | <p>Observable indicators: (1) Q1 answer correctly identifies the steepest segment and uses the word 'speed' or 'fast' — teacher listens during cold-calling. (2) Q3 answer describes or draws a horizontal line — teacher checks written work. (3) Race commentary (Q5) contains at least one correct graph-shape descriptor linked to a physical event (e.g., 'Runner B's line becomes less steep between 25 km and 35 km, showing they were slowing down'). (4) Prediction annotations exist in a third colour on every learner's original sketch — teacher scans during circulation.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | that... / I was surprised that...'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | — two differences you see between your prediction and your actual graph.' Cold-call one learner to share a surprise. Connect to phenomenon: 'So now look at your graph — where EXACTLY did Runner B lose the race? Point to it.' [Learners point.] 'That single moment is what our graph can show that Runner B's average speed — about 5.3 m/s — could NEVER show us.'                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Slope and intercept meaning from a table</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Position-Time Graphs Practice</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Learners return to the class Driving Question Board. Working in their pairs, they write on a sticky note or card (colour-coded: green = new understanding, orange = new question): one thing they can NOW explain about the marathon phenomenon that they could not explain before this lesson, and one new question the graph has raised for them. Pairs post their cards on the DQB in the appropriate columns: 'What we can now explain' and 'What we still need to find out.' The teacher reads three cards aloud to the class and briefly annotates the DQB, highlighting the cluster of questions around 'How do we find the exact speed from the graph?' as the bridge to Lesson 4. | Say: 'We now have a graph that tells the full story of the race. Update our Driving Question Board. Write: one thing you can now EXPLAIN — use the word graph in your answer. And one thing your graph has made you MORE curious about.' Give 4 minutes for writing and posting. Read three posted cards aloud — choose one strong 'explain' card, one weak/vague card (to model improving it), and one 'new question' card that foreshadows gradient. Say: 'Several of you asked HOW we calculate the exact speed from a graph — not just say it is steep, but get a number. That is exactly what we investigate in Lesson 4. Your graph is not finished — it is a TOOL we will keep using.' Point to the DQB column 'What we still need to find out' and say: 'These questions are our roadmap for the next five lessons.' | Driving Question Board update (knowledge-tracking protocol). By requiring learners to articulate both new understanding and new questions, the DQB update externalises the cumulative nature of the evidence-gathering sequence. The colour-coding (green/orange) allows the teacher and learners to see at a glance how the class's collective understanding is growing lesson by lesson, and sustains the motivating puzzle of the phenomenon across all 8 lessons. | Observable indicators: (1) Every pair has posted at least one green and one orange card. (2) 'Explain' cards reference graph shape or the marathon phenomenon specifically — not generic statements like 'I learned about graphs.' (3) New questions are answerable through future lessons (i.e., they are mathematically productive), not purely factual questions about the marathon event. |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Linear graphs word problem: cats</a><br><i>Related topic: not</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Learners open the cumulative model they began in Lesson 1 (a written/drawn model of 'how motion works' that they revise each lesson). Today's revision task: add the displacement-time graph of Runner A and Runner B to the model, with a key that links                                                                                                                                                                                                                                                                                                                                                                                                                                  | Say: 'Open your cumulative model from Lesson 1. Today you are adding the graphical layer. Stick or draw a small version of your displacement-time graph into your model. Then add a KEY — three entries: steep line = ?, gentle line = ?, horizontal line = ?. Fill in the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Cumulative model revision (progressive model building). The model acts as a personalised 'theory of motion' that grows across all 8 lessons. Adding the graph and the gradient sentence today makes the conceptual progression explicit: learners can                                                                                                                                                                                                                 | Observable indicators: (1) The model key correctly pairs steep/gentle/horizontal with faster/slower/stationary. (2) The gradient sentence, in the learner's own words, correctly captures the idea of rate of change of displacement (even if not using                                                                                                                                       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b> <a href="#">Average velocity and average speed from graphs</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>each type of line segment (steep, gentle, horizontal) to the physical meaning. Learners add one sentence in their own words: 'The gradient of a displacement-time graph tells us...' — they complete this sentence based on what they have reasoned today (they are not yet given the formal gradient formula; they write it in descriptive language, e.g., 'how quickly displacement is changing, which is the speed'). Learners share their completed sentence with a shoulder partner, then one sentence is chosen by the teacher to be written on a class anchor chart that will be added to each lesson.</p> | <p>physical meaning for each.' Give 5 minutes. Then: 'Now complete this sentence at the bottom of your model page: The gradient of a displacement-time graph tells us... — do not use the word gradient in your answer; explain it in plain language.' Give 2 minutes. Cold-call 2 learners to read their sentence. [10 seconds wait time each.] Choose the clearest sentence and write it on the anchor chart under 'Lesson 3 — Displacement-Time Graph.' Say: 'In Lesson 4 we will make this sentence more precise by calculating the gradient as a number. You have already discovered the idea — next lesson you will make it mathematical.' Collect squared-paper graphs for teacher review (return in Lesson 4).</p> | <p>literally see, by flipping back through their model, how their understanding has deepened from 'motion is distance and time' (Lesson 1) to 'motion is the shape of a graph' (Lesson 3) to, eventually, 'motion is the calculus of area and gradient' (Lessons 6–8). The unfinished sentence about gradient creates a productive anticipation for Lesson 4.</p> | <p>that terminology). (3) Learners who write 'the gradient tells us the distance' (confusing displacement with speed) are identified and their papers are flagged for targeted re-teaching at the start of Lesson 4. Teacher collects all graphs to check: correct scales, labelled axes, title, two colours, and crossing point circled.</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## D. TEACHER REFLECTION

After the lesson, consider: (1) PREDICTION QUALITY — Did the majority of learners predict that Runner B's line would be steeper initially and then flatten? If most drew two similar straight lines, it suggests Lesson 2's speed calculations were not yet connected to the idea of varying speed — revisit by opening Lesson 4 with a 5-minute recap of Runner B's checkpoint-by-checkpoint speeds. (2) SCALE CHOICE — How many pairs needed direct teacher intervention to choose a workable scale? If more than a third of pairs struggled, build a dedicated 5-minute 'scale-choice protocol' into the start of Lesson 4 before gradient work begins. (3) GRAPH-AS-NARRATIVE — Were learners able to write a race commentary (Q5) that used graph-shape language? If answers defaulted to 'Runner B was fast then slow,' without referencing line shape, the gradient-language bridge to Lesson 4 is not yet secure — use those written responses as worked examples at the start of Lesson 4. (4) DQB QUALITY — Did new questions cluster around 'how do we get a number from the steepness?' If so, this is the ideal launch point for Lesson 4. If questions were mostly about the marathon event itself (e.g., 'Who was Runner B?'), redirect the DQB in Lesson 4 by asking: 'Which of these questions can mathematics answer — and how?'

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                            |                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b> | Learners plotted the split-time data for Runner A and Runner B from the 2023 Nairobi Marathon on the same displacement-time graph using squared paper, two colours, labelled axes, and a title. They circled the point where Runner B's line crossed Runner A's line — the graphical moment that explains the race outcome.        |
| <b>What did I learn?</b>   | The shape of a displacement-time graph — not just its start and end points — tells the full story of motion. A steep line means high speed; a gentle line means low speed; a horizontal line means the runner is stationary. The gradient of the line at any section represents the speed of the runner at that stage of the race. |

**How does this explain the phenomenon?**

Using the completed dual displacement-time graph, learners explained why Runner B — despite being ahead at 30 km — lost the race: Runner B's line became noticeably less steep between 25 km and 35 km (slowing down), while Runner A's line maintained a consistent gradient throughout, overtaking Runner B during the slow section between 30 km and 35 km. A single average-speed calculation could never have revealed this — only the graph shape can.

## LESSON 4 (80 minutes): Interpreting Displacement-Time Graphs: Gradient as Velocity

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners interpret pre-drawn displacement-time graphs representing real-life Kenyan scenarios, calculate the gradient of each segment, and connect gradient to velocity — enabling them to read the full story of motion from a graph's shape.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Knowledge</b>            | Learners will know that: (a) the gradient of a displacement-time graph equals velocity; (b) a steeper positive gradient means higher velocity; (c) a zero gradient (horizontal segment) means the object is stationary; (d) a negative gradient means the object is returning towards its starting point; (e) average velocity over an interval is calculated as change in displacement ÷ change in time.                                                                                                                                                                                                                                                                |
| <b>Skills</b>               | Learners will be able to: (a) read displacement and time values from a multi-segment graph; (b) calculate the gradient of each segment and state its meaning in context; (c) identify stationary, constant-velocity, and return phases from graph shape; (d) construct a brief narrative of motion from a displacement-time graph; (e) use gradient calculations to compare velocities of two moving bodies.                                                                                                                                                                                                                                                             |
| <b>Attitudes</b>            | Learners will: (a) appreciate that a graph communicates richer information than a single number; (b) show integrity by reading graph values carefully and recording calculations honestly; (c) show responsibility by connecting mathematical results back to real-world meaning.                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Key Inquiry Question</b> | How do we use displacement-time graphs in real life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Purpose in Storyline</b> | In Lesson 3, learners drew displacement-time graphs from the Marathon split-time data and noticed that the shapes for Runner A and Runner B were different. In this lesson they move from drawing to systematic interpretation — calculating gradients segment by segment to quantify what each part of the graph reveals about velocity. By the end of the lesson, learners can fully answer Key Inquiry Question 2 and can update their Driving Question Board entry explaining exactly why Runner B was ahead at 30 km yet lost: his gradient (velocity) between 25 km and 35 km was lower than Runner A's near-constant gradient, and his late sprint came too late. |
| <b>Safety Notes</b>         | No physical hazards. Learners use rulers and pencils on printed graph sheets — remind learners to handle rulers carefully and not to lean on sharp edges. Ensure printed graph sheets are distributed before the lesson to avoid delays.                                                                                                                                                                                                                                                                                                                                                                                                                                 |

### B. LESSON OVERVIEW

Lesson 4 of 8 in the evidence-gathering sequence for Sub-Strand 2.9: Linear Motion. Learners receive a set of four pre-drawn displacement-time graphs on a printed worksheet. Each graph depicts a different Kenyan scenario: (1) a matatu travelling from Nakuru town centre, stopping at a stage, then continuing; (2) a marathon runner resting at a water station on Uhuru Highway; (3) a boda-boda rider cycling away from home and then returning; (4) an SGR Madaraka Express multi-stop journey between Nairobi and Mombasa. Using the Predict–Observe–Explain framework, learners first predict what each graph segment means, then systematically calculate gradients, then construct written motion narratives. The lesson culminates in DQB revision — learners update their board to answer "How do we use displacement-time graphs in real life?" — and a cumulative model revision linking gradient-as-velocity back to the Marathon phenomenon. The SGR data anchors the lesson in authentic Kenyan infrastructure and provides rich multi-segment gradient work.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
|-------|--------------------|---------------|----------------------|-------------------------------|
|-------|--------------------|---------------|----------------------|-------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Position vs. time graphs</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Average velocity and average speed from graphs</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners receive the printed worksheet containing four displacement-time graphs (no labels on the y-axis segments, only axes with scales). Working individually for 4 minutes, then sharing with a partner for 2 minutes, each learner writes answers to three prediction prompts printed at the top of the worksheet: (1) 'Look at Graph 1 — the matatu. Which segment do you think shows the matatu stopped at a stage? How can you tell just from the shape?' (2) 'In Graph 3 — the boda-boda — one segment slopes downward. What do you think that means about the rider's position?' (3) 'Without calculating anything, rank the four segments of Graph 4 (the SGR journey) from fastest to slowest. What visual clue did you use?' Learners write predictions in their exercise books before any instruction is given.</p> | <p>Distribute the worksheet face-down. Say: 'Do NOT turn it over until I say so.' After all sheets are distributed, say: 'Turn over. You have 4 minutes — write your predictions for the three questions at the top. I am not looking for right answers; I want your honest thinking right now.' Start a visible timer. Circulate silently — do not answer questions, only prompt with: 'Write something — even if you are not sure.' At 4 minutes, say: 'Now turn to your partner. Share your prediction for Question 2 only. You have 90 seconds.' After partner share, cold-call 3 pairs (choose one high-confidence, one mid-confidence, one uncertain based on circulation observations). For each, ask: 'What did your pair say — and what is your reasoning?' Allow 10 seconds WAIT TIME after each response before acknowledging. Do NOT correct errors yet. Record 2–3 key student predictions visibly on the board under the heading 'Our Predictions — Before Evidence'.</p> | <p>Strategy: Prediction Anchoring. By committing predictions to paper before instruction, learners activate prior intuitions about graph shape and motion. This creates a cognitive 'hook' — when the correct interpretation is later established through gradient calculation, the contrast with their prediction deepens understanding. Displaying predictions on the board also creates accountability: learners are motivated to check whether their intuition was right during the Observe phase.</p>                                                   | <p>Observable indicator: Every learner has at least one written prediction per prompt before partner share begins. Teacher scans for two specific misconceptions to address later: (a) learners who say a downward slope means 'slowing down' (confusing gradient sign with gradient magnitude — it means returning, not decelerating); (b) learners who rank SGR segments by length of segment rather than steepness. Note names of learners holding these misconceptions for targeted cold-calling in Phase 3.</p>                                                                      |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Position vs. time graphs</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Average velocity and average speed from graphs</a><br/> Source: Khan Academy (English</p>                                                                                                                                                    | <p>Learners work through the four graphs systematically in pairs using the structured 'Gradient Hunter' table printed on the reverse of their worksheet. The table has columns: Segment   Start point (t, s)   End point (t, s)   Change in displacement (<math>\Delta s</math>)   Change in time (<math>\Delta t</math>)   Gradient = <math>\Delta s / \Delta t</math>   Units   What this tells us about velocity. For Graphs 1–3, learners complete all segments (2–3 segments each). For Graph 4 (SGR Nairobi–Mombasa), they complete 5 segments representing: Nairobi departure → Athi River stop; Athi River → Emali</p>                                                                                                                                                                                                      | <p>Before releasing pairs to work, model the gradient calculation ONCE using a projected or board-drawn mini-graph (not one of the four on the worksheet). Say: 'I am going to show you the method using this example — a matatu travelling from Nairobi CBD to Westlands. Watch what I do.' Draw two points, read coordinates aloud, calculate <math>\Delta s</math> and <math>\Delta t</math>, divide, state units, write: "The matatu travelled at 0.8 km/min, which is 48 km/h." Then say: 'Now you do the same for your four graphs. Work in pairs — one person reads the coordinates, the other records and</p>                                                                                                                                                                                                                                                                                                                                                                   | <p>Strategy: Structured Data Table (Gradient Hunter). The pre-formatted table scaffolds the mathematical procedure into discrete retrievable steps, preventing learners from skipping to an answer without connecting <math>\Delta s</math> and <math>\Delta t</math>. Writing a one-sentence motion description for every segment forces translation between symbolic result and physical meaning — a key sensemaking move that connects mathematics to the Marathon phenomenon. The SGR multi-segment graph mirrors the Marathon split-time structure,</p> | <p>Observable indicators: (a) All pairs correctly record gradient = 0 with description 'stationary / not moving' for the horizontal SGR segment and the runner's rest segment. (b) At least one member of each pair can state the units of gradient as km/min or km/h without prompting. (c) Pairs correctly identify that the negative-gradient segment on the boda-boda graph means the rider is returning home (displacement decreasing), not slowing down. Teacher notes any pair still writing 'slowing down' for the negative segment — these pairs are prioritised in Phase 3.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>- US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>stop; Emali → Mtito Andei stop; Mtito Andei → Voi stop; Voi → Mombasa arrival. A horizontal segment (Mtito Andei rest stop, 10 minutes) is deliberately included. Learners read coordinates directly from the graph, substitute into the gradient formula, record units as km/min or km/h (conversion prompt provided), and write a one-sentence motion description for each segment. Time allocation: 18 minutes for pair work.</p>                                                                                                                                                                                                                                                             | <p>calculates. Swap roles after each graph.' Set a 18-minute timer. Circulate with purpose: (1) At 5 minutes — check that all pairs have correctly identified the horizontal segment on Graph 2 (runner resting) and have written gradient = 0; if not, ask 'What is the change in displacement during this segment?' and wait 10 seconds. (2) At 10 minutes — check Graph 4 Mtito Andei segment: ask 'What does a gradient of zero mean in the context of the SGR train?' (3) At 15 minutes — check units: prompt any pair writing 'km' instead of 'km/min' with: 'Gradient = <math>\Delta s \div \Delta t</math> — what are the units of each?' After 18 minutes, cold-call 4 pairs (one per graph) to share ONE gradient value and its motion meaning. Allow 10 seconds WAIT TIME per response. Write correct gradient values on board in a summary column.</p> | <p>making the connection explicit.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Average velocity for constant acceleration</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Instantaneous velocity and instantaneous speed from graphs</a><br/> <i>Related topic: not an exact match for this lesson.</i></p> | <p>Learners return to individual work for the 'Motion Narrator' task (12 minutes). Using their completed Gradient Hunter tables, each learner writes a full motion narrative for ONE assigned graph (teacher assigns each pair a graph by calling table numbers). The narrative must: (a) state the starting and ending displacement; (b) name each phase (e.g., 'constant velocity phase', 'stationary phase', 'return phase'); (c) quote the calculated gradient as the velocity in each phase; (d) compare two segments and explain which shows higher velocity and why. After individual writing (8 minutes), pairs join to form groups of four and compare narratives for the same graph —</p> | <p>Before individual writing begins, address the two key misconceptions identified in Phase 1 directly. Write on the board: 'CLAIM: A downward slope means the object is slowing down.' Ask the class: 'Agree or disagree? Use your gradient calculation from the boda-boda graph as evidence. 10 seconds to think.' Cold-call a learner who was noted holding this misconception. After their response, probe: 'What is the value of the gradient in that segment? What does the negative sign tell us about direction, not speed?' Then write the correct statement: 'A negative gradient means displacement is DECREASING — the object is moving back toward the origin.'</p>                                                                                                                                                                                   | <p>Strategy: Motion Narrator (Claim–Evidence–Reasoning writing). Translating a graph into a full written narrative requires learners to hold the mathematical result (gradient value) and its physical meaning (velocity) simultaneously, then sequence them into a coherent story. This is precisely the sensemaking required to answer the driving question. The group comparison step surfaces conflicting interpretations and requires evidence-based resolution — mirroring scientific argumentation practice (NGSS SEP 7: Engaging in Argument from Evidence). Connecting back to Runner A vs Runner B re-anchors the lesson in the phenomenon.</p> | <p>Observable indicators: (a) Each learner's written narrative contains at least one correctly quoted gradient value with units used as velocity. (b) Narratives correctly distinguish between 'stationary' (gradient = 0), 'constant velocity' (non-zero constant gradient), and 'returning' (negative gradient). (c) During the Marathon connection discussion, at least 3 cold-called learners can articulate that Runner A's steeper gradient while Runner B slowed (25–35 km) explains Runner A's win — linking graph interpretation directly to the phenomenon. Teacher uses a quick show-of-hands: 'Thumbs up if you can now explain, using the word gradient, why Runner A won' — target: ≥ 75% of class.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> <a href="#">for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>identifying agreements and resolving any differences in interpretation (4 minutes). Each group then nominates one speaker to present their narrative to the class in a 60-second 'pitch'. The class listens and can challenge with: 'Show me which segment you are describing on the graph.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>During individual writing, circulate and look for: learners who quote gradient correctly as velocity with units; learners who correctly use the phrase 'constant velocity' for straight-line segments. After group comparison and pitches, facilitate a whole-class discussion using the question: 'Runner B in our Marathon was ahead at 30 km. Looking at what we just learned about gradient — what would Runner B's displacement-time graph look like between 25 km and 35 km compared to Runner A's? Which graph shape wins the race?' Allow 15 seconds WAIT TIME. Cold-call 3 learners. Summarise on the board: 'Higher gradient while Runner B slows (25–35 km) = Runner A gains and wins.'</p>                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Driving Question Board (DQB) Creation</b><br/> <b>VIDEO:</b><br/> <a href="#">Linear graphs word problem: cats</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a><br/> <a href="#">for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Relating linear contexts to graph features</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a></p> | <p>Learners return to the class Driving Question Board displayed at the front. The board has been carrying sticky notes and questions from Lessons 1–3. Key Inquiry Question 2 — 'How do we use displacement-time graphs in real life?' — has been on the board since Lesson 1. Each learner now writes a new sticky note (green = answer, yellow = new question still open) and places it on the board. Their green note must answer KIQ 2 in 1–2 sentences using the word 'gradient'. Their yellow note (optional but encouraged) raises a new question — for example: 'What if the gradient is changing — the line is curved? Does that mean acceleration?' or 'Can we find the velocity at a single moment, not just over a segment?' The teacher then facilitates a 4-minute whole-class DQB review: reading 3–4 selected green notes aloud and 2–3 yellow notes. Yellow notes that</p> | <p>Distribute green and yellow sticky notes. Say: 'You now have evidence from four real Kenyan scenarios. On your GREEN note, answer this question — write it exactly as it appears on the board: How do we use displacement-time graphs in real life? Your answer must include the word gradient. On your YELLOW note, write something you are still wondering about — something today's lesson raised but did not answer. You have 3 minutes.' Set timer. Circulate — if any learner's green note omits 'gradient', tap the board and say quietly: 'Check the key word.' After notes are placed, read 3 green notes aloud without naming the author. After each, ask the class: 'Does this answer satisfy the question? Thumbs up or down.' For yellow notes, read 2 aloud and say: 'We are going to need these questions in Lesson 5. Hold onto those thoughts.' Point to</p> | <p>Strategy: Driving Question Board Revision (public knowledge-building). The DQB functions as a communal sensemaking artefact — it externalises the class's growing understanding and makes gaps visible. Moving a question from 'unanswered' to 'answered' by placing a green note is a motivating, metacognitive act. Yellow notes activate productive curiosity about curved graphs and instantaneous velocity, creating a natural bridge to Lessons 5–6 (velocity-time graphs and acceleration). This strategy reflects NGSS SEP 1 (Asking Questions) and SEP 6 (Constructing Explanations).</p> | <p>Observable indicators: (a) Every learner places at least one green note containing the word 'gradient' and a real-life context reference. (b) At least 40% of learners place a yellow note that references changing velocity, curves, or acceleration — indicating readiness for Lesson 5. (c) During the class vote on green notes, ≥ 80% of thumbs are up, indicating broad class consensus on the answer to KIQ 2. Teacher photographs the updated DQB board for reference in Lesson 5.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>anticipate velocity-time graphs or acceleration are explicitly affirmed: 'Excellent — that is exactly what we will need in Lesson 5.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>the still-unanswered DQB section: 'Look — our big driving question is still not fully answered. We can now explain what happened at every point of the race using gradient. But we still cannot fully explain the sprint — what happens when velocity is changing rapidly. That is our next investigation.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b> <a href="#">Slope and intercept meaning from a table</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b> <a href="#">Interpreting slope and y-intercept for linear models</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p> <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to their cumulative 'Motion Story Model' — a personal annotated diagram they have been building since Lesson 1. This model currently shows: a timeline of the Marathon with distance markers; an annotated split-time table distinguishing Runner A and Runner B; and a displacement-time sketch graph drawn in Lesson 3. In this lesson, learners add a new layer to their model: they annotate each segment of their Runner A vs Runner B displacement-time sketch graph with the calculated gradient (velocity) value, using coloured pens. They must label at least: the segment where Runner B leads (steeper gradient, higher velocity in km/min), the water-station segment (horizontal, gradient = 0), and the 25–35 km slowdown segment (comparing gradients of the two runners, showing Runner A's is steeper). Below their annotated graph, learners write one 'Model Claim' statement: 'I can now explain [part of the phenomenon] because I know that [gradient concept].' Learners who finish early are challenged to sketch what a curved displacement-time segment would look like and annotate it with a question.</p> | <p>Say: 'Open your Motion Story Model to the displacement-time sketch you added in Lesson 3. Today we are going to make it speak — we are going to annotate every segment with its gradient value and what that means about velocity.' Give 2 minutes for learners to retrieve their models. Display a model example (teacher's own annotated version projected or drawn on board) showing the annotation style: arrow pointing to a segment → gradient value → velocity interpretation in words. Say: 'Your model must show: the segment where Runner B is winning — label its gradient. The water-station stop — label its gradient. The 25–35 km slowdown — label both Runner A's gradient and Runner B's gradient and circle the winner.' Circulate for 8 minutes. Look for: correct gradient values transferred from the Gradient Hunter table; correct winner identification based on gradient comparison. After 8 minutes, cold-call 2 learners to read their 'Model Claim' statement aloud. After each, ask the class: 'Does this claim use the evidence we gathered today? Thumbs up or down.' Close the lesson by saying: 'Your model can now explain every straight-line phase of this race. But look at your graph — do the lines ever curve?'</p> | <p>Strategy: Cumulative Model Annotation (Progressive Model Revision). Annotating a previously drawn graph with newly calculated gradient values enacts NGSS SEP 2 (Developing and Using Models) — the model becomes more explanatory with each lesson. The 'Model Claim' writing exercise forces each learner to articulate the specific conceptual gain of this lesson in their own words, supporting metacognitive awareness of learning progression. Connecting back to the still-unexplained curved sprint segment creates purposeful incompleteness — a pedagogical strategy that sustains curiosity for Lesson 5.</p> | <p>Observable indicators: (a) Every learner's annotated graph shows at least three labelled gradient values with correct units and correct motion interpretation (stationary, constant velocity, or return). (b) Every learner correctly circles Runner A's 25–35 km segment as having the higher gradient (velocity) and writes this as the reason Runner A won. (c) 'Model Claim' statements reference gradient and velocity explicitly. (d) Extension learners sketch a curved segment and annotate it with a meaningful question — teacher notes these for use as a launch prompt in Lesson 5. Teacher collects 5–6 models at random for informal marking before Lesson 5.</p> |

|  |  |                                                                                                                                                                                                    |  |  |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  |  | In Lesson 5, we will discover what a curve on this graph means, and that will lead us to velocity-time graphs and acceleration — the final tools we need to explain the sprint finish completely.' |  |  |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

### D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Misconception check — Did learners consistently distinguish between a negative gradient (returning toward origin) and a decreasing gradient (decelerating)? If more than 25% of learners' green DQB notes or model annotations conflated these two ideas, plan a 5-minute re-teach at the start of Lesson 5 using the boda-boda graph. (2) Gradient calculation accuracy — Were learners reading coordinates precisely enough from the graph, or did rounding errors lead to significantly different gradient values across pairs? If so, consider adding a 'coordinate reading' mini-drill to Lesson 5's opening. (3) Phenomenon connection — Did learners successfully transfer gradient-as-velocity from the worksheet graphs back to the Marathon phenomenon during the DQB and model phases? If the Marathon connection felt forced rather than natural, review the transition question in Phase 3 and Phase 5. (4) DQB yellow notes — Were learners' yellow notes pointing meaningfully toward acceleration and velocity-time graphs? This is the key readiness indicator for Lesson 5. Count the yellow notes referencing 'changing velocity', 'curve', or 'acceleration' — target is  $\geq 40\%$  of class. (5) Pacing — The Observe phase (Gradient Hunter, 18 minutes) is the most time-intensive. If pairs did not complete Graph 4 (SGR), assign the remaining SGR segments as a 5-minute starter activity in Lesson 5 rather than cutting Phase 4 or Phase 5.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | Learners examined four pre-drawn displacement-time graphs showing Kenyan real-life motion scenarios: a matatu stopping at a Nakuru stage, a marathon runner resting at a water station on Uhuru Highway, a boda-boda rider returning home, and the SGR Madaraka Express making stops between Nairobi and Mombasa. Using the Gradient Hunter table, learners read coordinates from each segment and calculated gradient values for every phase of each journey.                                                                                                                          |
| <b>What did I learn?</b>                     | Learners established that the gradient of a displacement-time graph equals velocity; a zero gradient means the object is stationary; a negative gradient means the object is returning toward its starting point; and a steeper positive gradient means higher velocity. Learners applied this to the Marathon phenomenon by annotating Runner A and Runner B's sketch graphs with segment gradient values and identifying that Runner A's higher gradient while Runner B slowed (25–35 km) is the mathematical explanation for why Runner A won despite Runner B being ahead at 30 km. |
| <b>How does this explain the phenomenon?</b> | Learners explained that displacement-time graphs are used in real life to reveal the velocity at every phase of a journey — not just the overall average. By calculating the gradient of each segment, we can identify when a body was moving fast, slow, stationary, or returning. This answers Key Inquiry Question 2: we use displacement-time graphs to describe and compare motion in detail, segment by segment, which a single average speed calculation cannot do.                                                                                                              |

## LESSON 5 (40 minutes): Drawing Velocity-Time Graphs — Reading Acceleration in the Shape of the Race

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners construct velocity-time graphs from data tables and identify gradient as acceleration, building the mathematical tools needed to fully explain Runner B's dramatic sprint finish at the 2023 Nairobi Marathon.                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Knowledge</b>            | <ul style="list-style-type: none"> <li>– A velocity-time graph plots velocity (y-axis, m/s or km/h) against time (x-axis, seconds or minutes)</li> <li>– The gradient of a velocity-time graph equals acceleration: <math>a = \Delta v / \Delta t</math></li> <li>– A horizontal segment on a velocity-time graph represents constant velocity (zero acceleration)</li> <li>– A positive gradient means the object is accelerating (speeding up); a negative gradient means deceleration (slowing down)</li> <li>– The steeper the gradient, the greater the magnitude of acceleration</li> </ul> |
| <b>Skills</b>               | <ul style="list-style-type: none"> <li>– Draw an accurate velocity-time graph from a given table of values, including correct labelling of axes, units, scale, and title</li> <li>– Calculate the gradient of each segment of a velocity-time graph and interpret it as acceleration in <math>\text{m/s}^2</math></li> <li>– Identify and describe phases of uniform motion, acceleration, and deceleration from graph shape</li> </ul>                                                                                                                                                           |
| <b>Attitudes</b>            | <ul style="list-style-type: none"> <li>– Appreciate that a graph's shape — not just its end points — reveals the full story of motion, reflecting integrity in mathematical representation</li> <li>– Show responsibility by plotting data carefully and accurately, recognising that precision in mathematics mirrors precision in real-world measurement</li> </ul>                                                                                                                                                                                                                             |
| <b>Key Inquiry Question</b> | How do we use velocity-time graphs?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose in Storyline</b> | In Lessons 3 and 4 learners mastered displacement-time graphs and gradient as velocity. Lesson 5 advances the storyline by introducing velocity-time graphs, allowing learners to visualise and calculate acceleration directly — the missing mathematical tool needed to explain why Runner B's powerful but unsustained sprint in the final 2 km was not enough to beat Runner A's disciplined, consistent pace.                                                                                                                                                                                |
| <b>Safety Notes</b>         | No specific hazards. Ensure learners use rulers carefully when drawing axes and constructing lines on graph paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### B. LESSON OVERVIEW

This lesson marks a pivotal moment in the storyline sequence: learners have already calculated velocity and acceleration numerically (Lessons 1–2) and mastered displacement-time graphs (Lessons 3–4). Now they encounter a new type of graph — the velocity-time graph — which makes acceleration visible as the slope of a line rather than a single calculated number. The lesson is anchored in three complementary contexts: the 2023 Nairobi Marathon split-finish data (the anchoring phenomenon), a supporting ball-on-a-ramp scenario introduced briefly to provide a controlled physical reference, and a matatu journey on the Nakuru–Nairobi highway that provides a rich three-phase scenario (accelerating from a bus stop, cruising at constant speed, braking before Nakuru town) perfectly suited to constructing a multi-segment velocity-time graph.

Learners begin by predicting what a velocity-time graph for the matatu scenario would look like before they see any data, drawing on their intuitive understanding of motion and their prior work with displacement-time graphs. They then receive a printed data table for the matatu journey and construct the graph in pairs, identifying each segment's gradient and labelling it as acceleration or deceleration. The Explain phase connects the graph shape back to the marathon: Runner B's split data is re-examined as a velocity-time sketch, making viscerally clear how Runner B's steep positive gradient (sprint acceleration) in the final kilometres came too late to overcome Runner A's consistent, flat velocity profile over the whole race.

Throughout the lesson, the teacher uses targeted cold-calling, structured wait time, and a segment-annotation routine to build precision in mathematical language. By the end of the lesson, every learner should be able to state that the gradient of a velocity-time graph equals acceleration, sketch the three characteristic segment shapes (rising line, horizontal line, falling line), and connect these shapes to the real motion of runners and vehicles they know from their daily lives in Kenya.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                       | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Interpreting direction of motion from velocity-time graph</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Velocity-Time Graphs (Flexbook)</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | <p>Learners are presented with a written scenario — no numbers yet — describing a matatu leaving a bus stop on Uhuru Highway in Nairobi: it starts from rest, builds up speed over 10 seconds, travels at a steady speed for 30 seconds, then slows down and stops at the next stage. Each learner individually sketches what they think the velocity-time graph for this journey would look like on a blank axes grid in their exercise books, labelling the three phases. They write one sentence beneath the sketch predicting what the slope of each section will look like and what that slope might mean.</p> | <p>Display the matatu scenario on the board (written in large text or on a manila card). Say: 'Before we look at any numbers, I want you to think about what is actually happening to the matatu's speed at each stage of this journey. In your exercise books, sketch what you think a velocity-time graph would look like — velocity on the y-axis, time on the x-axis. You have 3 minutes working silently and individually.' Start a visible timer. After 3 minutes, say: 'Now look at your sketch. Under it, write ONE sentence describing what the slope of each section means about the matatu's motion.' Allow a further 2 minutes. After 5 minutes total, cold-call 3 learners to describe their sketches aloud — choose one learner whose sketch shows all three phases correctly, one who has the acceleration phase correct but the deceleration phase flat, and one who has drawn a straight diagonal line the whole way. For each, ask: 'What did you decide to do with the line during the steady-speed phase? Why?' Apply a minimum 12-second wait time after each question before accepting an answer. Do NOT correct errors yet — note the range of predictions on the board as three labelled sketches titled 'Prediction A', 'Prediction B',</p> | <p>Activate-and-Anchor Prediction: By committing to a sketch before seeing data, learners activate prior knowledge from displacement-time graphs (Lessons 3–4) and create a cognitive 'hook' that makes the subsequent data observation more meaningful. Recording predictions publicly creates accountability and a reference point for revision, embodying the NGSS Science and Engineering Practice of developing and using models.</p> | <p>Teacher circulates during the 5-minute prediction window and looks for three indicators: (1) Does the learner start the graph at <math>v = 0</math> (matatu begins from rest)? (2) Does the learner draw a rising line, a horizontal line, and a falling line — or some other combination? (3) Does the learner's written sentence use the word 'slope', 'gradient', 'speed', or 'acceleration' in a mathematically meaningful way? Learners who draw a horizontal line throughout have not yet transferred gradient-as-rate from displacement-time graphs — the teacher notes their names and targets them during Observe Phase pair work.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'Prediction C' to return to in the Explain phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Interpreting change in speed from velocity-time graph</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Velocity-Time Graphs (Flexbook)</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Learners receive a printed data table (or copy it from the board) showing the velocity of the matatu at regular time intervals during its journey from the Kencom bus stop on Moi Avenue, Nairobi. The table shows: t = 0 s, v = 0 m/s; t = 2 s, v = 3 m/s; t = 4 s, v = 6 m/s; t = 6 s, v = 9 m/s; t = 8 s, v = 12 m/s; t = 10 s, v = 15 m/s; t = 20 s, v = 15 m/s; t = 30 s, v = 15 m/s; t = 40 s, v = 15 m/s; t = 46 s, v = 9 m/s; t = 52 s, v = 3 m/s; t = 55 s, v = 0 m/s. Working in pairs, learners plot these points on graph paper using a provided scale (x-axis: 1 cm = 5 s; y-axis: 1 cm = 3 m/s), draw straight lines connecting points within each segment, and label the three phases: 'Phase 1: Accelerating', 'Phase 2: Constant velocity', 'Phase 3: Decelerating'. They then calculate the gradient of Phase 1 and Phase 3 using rise-over-run and record the values with units. | Distribute graph paper and the data table simultaneously. Say: 'You are going to construct a velocity-time graph for this matatu journey. Work in pairs. Step one — decide on your scale and label your axes with units. Step two — plot every point in the table. Step three — draw straight lines connecting points within each of the three phases. Step four — label the three phases. You have 10 minutes.' Circulate actively. At the 4-minute mark, pause the class for 30 seconds and say: 'Quick check — raise your hand if you have already labelled both axes with the correct units. Good. If you haven't, do that now before plotting any more points — units are not optional in mathematics.' At the 8-minute mark, when most pairs have completed the graph, say: 'Now, without touching your calculator, estimate the gradient of Phase 1. Talk to your partner for 30 seconds.' After 30 seconds, cold-call 2 pairs: 'Pair at the back — what gradient did you estimate for Phase 1, and how did you calculate it?' Apply 12-second wait time. Then cold-call a second pair: 'Do you agree? What value did you get, and what are the units of this gradient?' Confirm on the board: gradient of Phase 1 = $(15 - 0) \div (10 - 0) = 1.5 \text{ m/s}^2$ . Write clearly: 'Gradient of v-t graph = acceleration ( $\text{m/s}^2$ ). Repeat the gradient calculation routine for Phase 3 with a third cold-call pair. | Data-to-Graph Construction with Segment Annotation: Learners do not merely observe a pre-drawn graph — they construct it from raw data, which deepens understanding of what each plotted point represents. The segment-annotation routine (labelling phases, calculating gradients) is a disciplined sensemaking practice that connects visual shape to numerical meaning, enacting the NGSS practice of Analysing and Interpreting Data. The pair structure ensures peer scaffolding during the technical plotting task. | Teacher checks three specific quality indicators while circulating: (1) Are axes labelled with both quantity name AND units (e.g., 'Velocity (m/s)' and 'Time (s)')? — this is a non-negotiable mathematical standard. (2) Are points plotted accurately and connected with a ruler (straight-line segments, not a freehand curve)? (3) Is the gradient calculation set out as $\Delta v/\Delta t$ with correct units of $\text{m/s}^2$ ? Pairs who attempt to draw a smooth curve through all points have confused velocity-time with displacement-time graph conventions — teacher intervenes immediately with: 'Each segment represents a different type of motion. Should they be curves or straight lines? Why?' |
| <b>Explain Phase</b><br> <b>VIDEO:</b><br><a href="#">Interpreting</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Learners engage in a structured whole-class discussion guided by three focus questions written on                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Return to the three prediction sketches on the board. Say: 'Look at Prediction A, B, and C. Your                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Predict-Observe-Explain (POE) Closure with Phenomenon Reconnection: Returning to the                                                                                                                                                                                                                                                                                                                                                                                                                                      | Teacher collects or scans (walks around to read) the two-sentence written explanations. Look for                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">direction of motion from velocity-time graph</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b></p> <p><a href="#">Uniform Acceleration (Flexbook)</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: CK-12</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>the board. First, they compare their completed velocity-time graph with the three 'Prediction' sketches saved from the Predict phase — identifying which prediction was most accurate and why. Second, they apply their gradient-calculation skill to a new context: the teacher displays a simplified velocity-time sketch for Runner B at the 2023 Nairobi Marathon (derived from the split-time data introduced in Lesson 1), showing Runner B's velocity dropping from approximately 5.6 m/s (about 20 km/h) to 4.9 m/s (17.6 km/h) between km 25 and km 35, then surging back to 6.2 m/s in the final 2 km (km 40–42). Learners calculate the gradient (acceleration) for each segment and annotate their marathon data sheet. Third, learners write a two-sentence explanation in their exercise books: 'The gradient of a velocity-time graph tells us _____. This explains what happened to Runner B between km 35 and km 42 because _____.'</p> | <p>real graph is now in front of you. Which prediction was closest to the truth — and more importantly, which part of the prediction was wrong and why?' Cold-call the learner whose sketch was used for 'Prediction C' (the straight diagonal line): 'Looking at your prediction now, what would a straight diagonal line across the whole graph have meant about the matatu's motion?' Apply 15-second wait time. Accept and build on the answer. Then transition: 'We have been thinking about a matatu. Let us go back to our marathon.' Project or draw the Runner B velocity-time sketch on the board. Say: 'This is a velocity-time sketch for Runner B, constructed from the split-time data we have been using since Lesson 1. I want you to calculate the gradient between km 35 and km 42 — remember, time values are on the x-axis. Work individually for 3 minutes.' After 3 minutes, cold-call 2 learners for their gradient values and units. Write: 'Runner B's acceleration in the final sprint = +0.027 m/s<sup>2</sup>.' Then ask: 'If Runner B was accelerating, why did Runner A still win?' Apply 15-second wait time. Cold-call 3 learners. Guide toward: 'Because Runner B's average velocity over the full race was lower — the gradient being positive in the last section does not undo the low velocity in the middle section.' Conclude by having all learners complete the two-sentence written explanation in their books.</p> | <p>saved predictions makes the Explain phase a genuine revision of thinking rather than a passive reception of information. The two-sentence written explanation is a 'language frame' sensemaking strategy — the sentence stems require learners to state the mathematical principle AND connect it explicitly to the anchoring phenomenon, practising the NGSS Science and Engineering Practice of Constructing Explanations. Writing individually after whole-class discussion ensures every learner consolidates the idea in their own words.</p> | <p>three quality indicators: (1) The first sentence correctly names gradient as acceleration and states the unit (m/s<sup>2</sup>). (2) The second sentence correctly identifies that Runner B's sprint acceleration in the final 2 km was positive but that his overall velocity profile was lower than Runner A's. (3) The explanation uses at least one number from the graph (e.g., a specific gradient value or velocity). Learners whose second sentence merely says 'Runner B was fast at the end' without mathematical reference have not yet connected the graph shape to the quantitative explanation — teacher flags these for targeted follow-up in the DQB phase.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Learners revisit the class Driving Question Board displayed at the front of the room. Each pair is given one sticky note (or writes in</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Direct learners' attention to the DQB. Say: 'Our driving question asks: how can we use mathematics — calculations AND</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Claim-Evidence-Question Board Sorting: Organising learner contributions into 'can explain', 'still wonder', and 'new connections'</p>                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Teacher scans the sticky notes / written DQB contributions for two indicators: (1) Are 'confident claims' genuinely supported by</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p> <b>VIDEO:</b><br/> <a href="#">Interpreting change in speed from velocity-time graph</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Uniform Acceleration Practice</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>a designated DQB section of their exercise book if sticky notes are unavailable). They write one of three things: (a) a new question that today's lesson has raised for them about velocity-time graphs or acceleration, (b) a claim they can now make confidently about the marathon phenomenon with supporting evidence from today's graph work, or (c) a connection between today's lesson and a real-world Kenyan context beyond the marathon (e.g., a boda boda accelerating from a junction in Kisumu, a train leaving Nairobi terminus, a boat on Lake Victoria). Pairs share their sticky note contribution with the class in a 3-minute rapid round before placing it on the board.</p> | <p>graphs — to fully describe the motion of any moving body? In Lessons 3 and 4 we answered part of this question using displacement-time graphs. Today we have added a second mathematical tool. I want you to think: what can we now say, and what can we still NOT yet say?' Distribute sticky notes or direct learners to the DQB section of their books. Say: 'With your partner, decide whether you are adding a NEW QUESTION, a CONFIDENT CLAIM, or a NEW CONNECTION. You have 2 minutes to write it.' After 2 minutes, do a rapid round: each pair reads their contribution aloud in one sentence — no more. Teacher organises contributions into three columns on the DQB as they are shared: 'We can now explain...', 'We still wonder...', 'New connections'. After the round, highlight two or three 'We still wonder' questions that will be addressed in Lesson 6 (interpreting velocity-time graphs: area under the graph as distance). Say: 'Hold onto these questions — they are exactly what we will attack next lesson.'</p> | <p>columns makes the structure of scientific reasoning visible — learners see that good mathematics learning generates new questions as well as answers. This strategy enacts the NGSS practice of Engaging in Argument from Evidence by requiring learners to distinguish between claims they can support with today's graph evidence and genuine open questions. The public display maintains a cumulative record of the class's growing understanding across the eight-lesson sequence.</p> | <p>evidence from today's work (e.g., 'We can calculate Runner B's acceleration in the sprint finish as <math>+0.027 \text{ m/s}^2</math> from the gradient of the velocity-time graph') rather than vague statements? (2) Are 'new questions' mathematically productive — pointing toward content still to come (e.g., 'Can we find the total distance from a velocity-time graph?') rather than off-topic? Vague claims or absent questions signal that the learner has not yet consolidated the lesson's core idea and need individual follow-up.</p> |
| <p><b>Model Building Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Acceleration vs. time graphs</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p>                                                                                                                                                                                                                                                                                                             | <p>Learners retrieve their cumulative motion model — the annotated sketch or diagram of the marathon race that they have been building and revising since Lesson 1. Today they add a new layer: alongside their existing displacement-time graph sketches for Runner A and Runner B (constructed in Lesson 3), they now add small velocity-time sketch graphs for each runner, drawn directly beneath the corresponding displacement-time graph so the</p>                                                                                                                                                                                                                                          | <p>Say: 'Get out your cumulative marathon model from your folder or the back of your exercise book. You have been building this since Lesson 1. Today we are adding a second layer of graph evidence.' Write on the board: 'Task: Add velocity-time sketches for Runner A and Runner B. Annotate with at least 2 gradient calculations. Write the caption: Displacement-time graph shows ____ / Velocity-time graph shows _____. Add the arrow: Runner A wins because ____.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Cumulative Model Revision with Dual-Graph Alignment: Placing the velocity-time graph directly beneath the displacement-time graph for the same time axis is a disciplined representational strategy that builds multi-representational fluency — a core competency in the KICD curriculum. The vertical alignment makes visible the relationship between position, velocity, and acceleration in a single visual structure, enacting the NGSS</p>                                           | <p>Teacher checks three model-quality indicators: (1) Are the velocity-time sketches drawn on a correctly labelled set of axes (velocity vs. time, not distance vs. time)? (2) Do the gradient annotations include both a numerical value and the unit <math>\text{m/s}^2</math> (not just a slope description)? (3) Does the 'Runner A wins because...' sentence reference the velocity-time graph specifically — i.e., does it mention gradient, average velocity, or a specific</p>                                                                  |

|                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                   |                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p> <b>HTML:</b><br/> <a href="#">Uniform Acceleration (PLIX)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>two graph types are vertically aligned for the same time axis. Learners annotate the velocity-time sketches with at least two gradient calculations (one acceleration, one deceleration) and write a two-phrase caption: 'Displacement-time graph shows... Velocity-time graph shows...' They also add an arrow labelled 'Runner A wins because...' connecting the two velocity-time sketches, completing in words and numbers why Runner A's flatter, more consistent velocity-time profile produced a lower total time despite Runner B's higher peak velocity.</p> | <p>Allow 5 minutes for individual model revision. Circulate and look specifically at the arrow annotation. After 5 minutes, cold-call 2 learners to read their 'Runner A wins because...' sentence aloud. Apply 12-second wait time after each. If a learner says only 'because he was faster', probe: 'Faster where? At which kilometre? What does the gradient of his velocity-time graph tell us about his acceleration compared to Runner B?' Conclude the lesson by saying: 'Your model now has two types of graph for the same motion. Next lesson we will discover a third piece of information hidden in the velocity-time graph — the area under the line. Think about what that might represent before our next class.'</p> | <p>practice of Developing and Using Models. The 'Runner A wins because...' annotation requires learners to synthesise all five lessons' worth of learning into a single evidence-based claim.</p> | <p>segment — rather than restating the finishing time alone? Models that lack numerical gradient annotations are flagged: the learner has grasped the visual concept but not yet the quantitative skill, which is the target outcome of Lesson 5.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## D. TEACHER REFLECTION

1. During the Predict phase, how many learners spontaneously drew three distinct segments (rising, horizontal, falling) versus a single diagonal line — and what does this distribution tell me about how well the displacement-time graph work from Lessons 3 and 4 transferred to the new graph type?
2. When learners calculated the gradient of Phase 1 of the matatu graph, did they correctly attach the unit  $\text{m/s}^2$  without prompting, or did they omit units or write  $\text{m/s}$ ? What targeted intervention will I use at the start of Lesson 6 to address unit precision?
3. In the Explain phase, how many learners' two-sentence written explanations connected Runner B's positive gradient in the sprint finish to his overall lower average velocity — versus those who simply described the shape of the graph without a quantitative claim? What does this tell me about the depth of understanding of the relationship between local gradient and overall motion?
4. Reviewing the DQB contributions, how many pairs generated a 'new question' that points toward the area-under-the-graph concept (e.g., 'Can we find distance from a velocity-time graph?')? If fewer than one-third of pairs reached this question independently, how will I bridge to that concept at the start of Lesson 6?
5. In the Model Building phase, did learners' vertical alignment of displacement-time and velocity-time graphs reveal any misconceptions about the relationship between the two — for example, did any learner draw the velocity-time graph with the same shape as the displacement-time graph? How will I address this misconception before Lesson 6?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>What did I observe?</b></p> | <p>Learners observed that a velocity-time graph for a matatu journey on Uhuru Highway, Nairobi, produces three visually distinct segments: a rising straight line (acceleration from the Kencom bus stop), a horizontal straight line (constant cruising velocity), and a falling straight line (braking before the next stage). They also observed Runner B's velocity-time sketch derived from marathon</p> |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | split data, showing a mid-race dip in velocity followed by a steep positive gradient in the final 2 km sprint.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>What did I learn?</b>                     | The gradient of a velocity-time graph equals acceleration ( $a = \Delta v / \Delta t$ , measured in $\text{m/s}^2$ ). A horizontal segment means zero acceleration (constant velocity); a positive gradient means the object is speeding up; a negative gradient means it is decelerating. The shape of the velocity-time graph — its slopes, not just its start and end points — quantifies how rapidly velocity is changing at every moment of the journey.                                                                                                                                                                   |
| <b>How does this explain the phenomenon?</b> | The velocity-time graph of the Nairobi Marathon now mathematically explains what the split-time tables could only hint at: Runner B's steep positive gradient between km 35 and km 42 confirms a genuine sprint-finish acceleration, but the graph also shows that his velocity during km 25–35 was significantly lower than Runner A's flat, consistent velocity profile across the same period. The gradient calculations make it possible to state, with numbers and units, exactly how much Runner B accelerated — and why, despite that acceleration, his overall average velocity was lower, giving Runner A the victory. |

## LESSON 6 (80 minutes): Interpreting Velocity-Time Graphs: Area Under the Graph as Distance

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | To enable learners to draw, read, and interpret velocity-time graphs, extract acceleration and deceleration from gradient, and calculate total distance travelled using area under the graph (counting squares, trapezium rule, and rectangle method) — connecting these skills to explain why Runner B lost the 2023 Nairobi Marathon despite leading at 30 km.                                                                                                                                                                  |
| <b>Knowledge</b>            | Learners will know that: (i) the gradient of a velocity-time graph gives acceleration (positive gradient) or deceleration (negative gradient); (ii) a horizontal segment on a velocity-time graph represents constant velocity and zero acceleration; (iii) the area under a velocity-time graph equals total distance (displacement) travelled; (iv) area can be calculated by counting grid squares or by decomposing regions into rectangles and trapeziums.                                                                   |
| <b>Skills</b>               | Learners will be able to: (i) draw a velocity-time graph from a given data table; (ii) calculate acceleration and deceleration from the gradient of line segments; (iii) calculate total distance travelled by finding the area under the graph using rectangles, trapeziums, and square-counting; (iv) compare the distance covered by two moving bodies in the same time interval using their velocity-time graphs.                                                                                                             |
| <b>Attitudes</b>            | Learners will: (i) appreciate that a graph shape — not just start and end points — tells the full story of motion; (ii) show integrity when reading values accurately from graphs; (iii) collaborate responsibly in group calculations; (iv) connect mathematical tools to real-life Kenyan contexts such as marathon running, matatu journeys, and SGR train travel.                                                                                                                                                             |
| <b>Key Inquiry Question</b> | How do we use velocity-time graphs?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose in Storyline</b> | In Lessons 4–5 learners mastered displacement-time graphs and average speed. Lesson 6 introduces the velocity-time graph as the next evidence layer: learners discover that the shape of Runner B's velocity-time graph — specifically the area lost under the curve between 25 km and 35 km — mathematically explains why Runner A overtook and won. This lesson directly resolves the central puzzle of the phenomenon and equips learners to update the Driving Question Board entry for 'How do we use velocity-time graphs?' |
| <b>Safety Notes</b>         | No physical hazards. Learners use rulers and pencils on graph paper; remind them to handle geometrical sets carefully. Ensure adequate spacing between desks during group work to prevent crowding.                                                                                                                                                                                                                                                                                                                               |

### B. LESSON OVERVIEW

Learners begin by predicting what a velocity-time graph for each marathon runner might look like, sketching rough curves from intuition. They then receive printed velocity-time data tables for two real-world Kenyan contexts — an SGR Madaraka Express journey (Nairobi–Mombasa segment) and a 400 m school race at Nakuru High School — and plot and interpret these graphs carefully. Through guided sensemaking, learners discover how to extract acceleration from gradient and total distance from area under the graph using rectangles, trapeziums, and square-counting. They apply this understanding retrospectively to Runner A and Runner B's velocity-time graphs, computing the area under each curve for the 25–35 km segment to prove mathematically why Runner B lost. The lesson closes with a DQB update and a revised cumulative model showing velocity-time graphs as a third motion-description tool alongside distance-time and displacement-time graphs.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
|-------|--------------------|---------------|----------------------|-------------------------------|
|-------|--------------------|---------------|----------------------|-------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Interpreting change in speed from velocity-time graph</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Velocity-Time Graphs (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners receive a blank velocity-time axes (velocity on y-axis, time on x-axis) and the verbal description of Runner A and Runner B from the phenomenon. Working individually for 4 minutes, they sketch what they think each runner's velocity-time graph looks like — no calculation, pure intuition. They annotate their sketches with labels such as 'sprinting here', 'slowing at water station', 'steady pace'. They then share with a shoulder partner and note one similarity and one difference between their two sketches. A class gallery of 6 volunteer sketches is posted on the board for reference throughout the lesson.</p> | <p>Distribute blank axes. Say: 'Before we do any maths, I want you to draw what you think Runner A and Runner B's velocity-time graphs look like — use what you already know about their race from Lesson 1. You have 4 minutes. Work alone first.' [START TIMER. Circulate silently for 2 minutes, noting misconceptions — e.g. learners drawing horizontal lines for both runners, or confusing velocity-time with displacement-time shape.] After 4 minutes: 'Turn to your shoulder partner. Tell them: one thing your graphs have in common, and one thing that is different. You have 90 seconds.' [WAIT. Circulate.] Cold-call 3 pairs to share. Probe: 'Why did you make Runner B's graph go higher at the start?' and 'What does a flat line on a velocity-time graph mean — is the runner stopped or moving at steady speed?' [WAIT TIME: 12 seconds after each question before taking responses.] Post 6 volunteer sketches on the board. Say: 'We will come back to these at the end of the lesson to see who was closest.'</p> | <p>Elicitation of Prior Knowledge (Predict-first protocol): By committing a sketch to paper before instruction, learners activate and externalise their intuitive models of velocity change. This surfaces misconceptions — particularly confusion between the shape of a displacement-time graph and a velocity-time graph — so the teacher can address them explicitly during the Observe phase. The shoulder-partner share encourages learners to articulate their reasoning, deepening initial engagement with the phenomenon.</p> | <p>Teacher circulates and looks for: (i) learners who draw a horizontal line for both runners — indicates confusion between constant position and constant velocity; (ii) learners who draw Runner B's graph as continuously descending — indicates confusion between deceleration and negative velocity; (iii) learners who draw the y-axis as 'distance' — indicates prior lesson vocabulary has not consolidated. These learners are targeted for proximity support during Phase 2.</p>                                                    |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Average velocity for constant acceleration</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Velocity-Time Graphs (Flexbook)</a><br/> Source: CK-12</p>                                                                                                                                              | <p>Learners work in groups of four to plot and read two velocity-time graphs from printed data tables. GRAPH 1 — SGR Madaraka Express (Nairobi to Athi River segment, approximately 30 minutes): the table gives velocity in km/h at 5-minute intervals: 0, 60, 120, 150, 150, 130, 90, 0. Learners plot this on a 1 cm = 10 minutes (x) and 1 cm = 20 km/h (y) grid. GRAPH 2 — 400 m school race at Nakuru High School: the table gives velocity in m/s at 2-second intervals over 60 seconds for two athletes, Athlete Achieng</p>                                                                                                             | <p>Distribute data tables, graph paper, and observation worksheets. Say: 'In your groups you are going to plot two real velocity-time graphs — one for the SGR train on its way out of Nairobi, and one for two athletes in a 400 m race at Nakuru High School. Plot carefully — use a ruler, label your axes with units, and give each graph a title.' [Allow 12 minutes for plotting. Circulate and check: axes labelled with units, correct scale chosen, points joined with straight lines between consecutive points — not curves</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Data-to-Graph Translation (Structured Observation): Physically plotting data points and joining them forces learners to engage with each value in the table, making the shape of motion visible rather than abstract. The two-context design (train journey and athletics race) provides redundancy — if a learner does not connect with one context, the other reinforces the same concept. The structured observation worksheet scaffolds attention to phases of motion before any calculation, building the conceptual</p>       | <p>Teacher checks for: (i) correctly labelled axes with units (velocity in km/h or m/s on y-axis; time on x-axis) — a missing unit is flagged immediately; (ii) straight-line segments drawn between data points (not freehand curves) — groups drawing smooth curves are redirected: 'We join consecutive data points with straight lines because we have data only at those intervals'; (iii) observation worksheet entries that use the words 'acceleration', 'constant', and 'deceleration' correctly in context — groups using vague</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>(steady pace) and Athlete Kamau (fast start, tires midway, sprints last 10 s). Learners plot both athletes on the same axes using different colours. From each graph learners complete a structured observation worksheet: (a) identify and label phases — acceleration, constant velocity, deceleration; (b) describe the shape of the graph during each phase; (c) record the steepness (gradient) of the acceleration and deceleration segments; (d) describe what is happening to the moving body during each phase in plain language. Groups record observations on A3 paper for sharing.</p>                                                                                                | <p>— for this piecewise linear data.] After plotting, say: 'Now fill in your observation worksheet. For each graph, identify the phases and describe the shape. Do not calculate yet — just describe what you see.' [Allow 6 minutes.] After worksheets are partially completed, pause the class. Cold-call one group: 'What shape is the SGR graph during the first 10 minutes — and what is the train doing during that time?' [WAIT TIME: 12 seconds.] Cold-call a second group: 'On the Nakuru race graph, whose line is steeper at the start — Achieng's or Kamau's — and what does that steepness tell us?' [WAIT TIME: 12 seconds.] Affirm and clarify: 'A steeper slope on a velocity-time graph means velocity is changing faster — that means greater acceleration or greater deceleration.' Write this on the board as a class anchor statement.</p> | <p>vocabulary (acceleration phase, constant velocity phase, deceleration phase) needed for Phase 3 sensemaking.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>language ('going fast', 'slowing') are prompted: 'Can you use the word acceleration or deceleration in that sentence?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Why distance is area under velocity-time line</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Average velocity and average speed from graphs</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a></p> | <p>The class reconvenes for direct instruction and guided practice in two parts. PART A — Gradient = Acceleration: The teacher demonstrates on the SGR graph (drawn large on the board) how to calculate the gradient of the acceleration phase: gradient = change in velocity ÷ change in time = (120 – 0) km/h ÷ (10 – 0) min, converted to m/s<sup>2</sup>. Learners calculate the deceleration phase gradient independently, then check with their group. They confirm that a negative gradient means deceleration. PART B — Area Under Graph = Distance: The teacher introduces the key concept: 'The area of the region between the velocity-time graph and the time axis equals the total</p> | <p>PART A: Draw the SGR velocity-time graph large on the board. Say: 'The gradient of a velocity-time graph tells us how fast velocity is changing — that is acceleration. Watch how I calculate the gradient for the acceleration phase.' [Model calculation step by step, writing units at every step.] Say: 'Now you calculate the gradient of the deceleration phase — the segment where the train slows to zero. Work individually, then compare with your group. You have 3 minutes.' [WAIT. Circulate. After 3 minutes, cold-call 2 learners — choose one confident and one who appeared uncertain during circulation — to share their answers.] [WAIT TIME: 12</p>                                                                                                                                                                                      | <p>Shape Decomposition and Area Calculation (Quantitative Sensemaking): Decomposing the area under a velocity-time graph into triangles, rectangles, and trapeziums connects new content (area as distance) to prior geometric knowledge, making the concept accessible. The deliberate pairing of shape-decomposition and square-counting as independent verification methods builds mathematical confidence and self-checking habits. The marathon application at the end of Phase 3 creates a direct, emotionally resonant payoff — the same Kenyan athletes learners have been tracking since Lesson 1 — making the abstract area calculation meaningful and</p> | <p>Teacher checks for: (i) correct identification of shapes (triangle, rectangle, trapezium) within the graph region — learners who shade the entire rectangle up to the peak velocity are redirected to shade only to the line; (ii) correct formula application — trapezium area = <math>\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}</math>, with correct identification of which dimensions are 'height' and which are 'parallel sides' in the context of the graph; (iii) correct units for area — velocity (m/s or km/h) × time (s or h) = distance (m or km) — learners who write 'm/s<sup>2</sup>' for the area answer are prompted: 'What are the units of the two axes? What do you get when you multiply them?'; (iv) the marathon comparison calculation: both area</p> |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |                                                                                                        |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------|
| <p>for similar readings</p> | <p>distance travelled.' The teacher works through the SGR graph, decomposing it into recognisable shapes: a triangle (acceleration phase), two rectangles (constant velocity phase), and a trapezium (deceleration phase), and calculates area for each. Learners then independently calculate the total area under Athlete Kamau's velocity-time graph for the Nakuru 400 m race, using both shape decomposition and square-counting as a check. Finally, the teacher presents a simplified version of the Runner A vs Runner B velocity-time graph for km 25–35 of the Nairobi Marathon. Learners calculate the area under each curve for this segment. The calculation reveals that Runner B's area (distance covered) in km 25–35 is smaller than Runner A's — proving mathematically that Runner A overtook Runner B after his slowdown even though they covered the same total 42 km race distance.</p> | <p>seconds between nomination and response.] Correct errors. Write on board: 'Gradient of velocity-time graph = acceleration (<math>\text{m/s}^2</math>). Positive gradient = acceleration. Negative gradient = deceleration. Zero gradient = constant velocity.' PART B: Say: 'Now the big idea of today's lesson. Look at the SGR graph. The area of the region under the graph — between the line and the time axis — equals the total distance the train travelled. Let me show you how to find it by breaking the shape into pieces we know.' [Decompose into triangle, rectangles, trapezium. Calculate each. Sum them.] Say: 'Now use the same method to find the total distance Athlete Kamau ran. Use shape decomposition first, then count squares to check. You have 8 minutes.' [Circulate actively. Prompt groups who are stuck: 'What shapes can you see when you look at the graph?'] Cold-call one group to present their answer and method on the board. [WAIT TIME after each board explanation: 10 seconds for peer questions.] MARATHON APPLICATION: Display the simplified Runner A vs Runner B velocity-time graph for km 25–35. Say: 'Calculate the area under Runner A's line for this 10 km segment. Then calculate the area under Runner B's line. What do you notice?' [Allow 6 minutes.] Cold-call 2 groups. Say: 'So — mathematically — who covered more distance in the final 12 km? Why did Runner A win?' [WAIT TIME: 15 seconds. Expect: 'Runner A covered more distance in km 25–35 because the area under Runner A's graph is larger, meaning Runner A's velocity was</p> | <p>memorable.</p> | <p>values computed and a clear written conclusion linking larger area to greater distance covered.</p> |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | higher for longer.']                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b> <a href="#">Interpreting change in speed from velocity-time graph</a></p> <p>Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b> <a href="#">Instantaneous velocity and instantaneous speed from graphs</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Groups spend 6 minutes updating the class Driving Question Board. Each group receives three sticky notes in different colours: ORANGE — 'What we can now answer'; GREEN — 'New questions this lesson raised'; PURPLE — 'What we still need to find out to fully answer the driving question'. Groups add their notes to the DQB column labelled 'How do we use velocity-time graphs?' Representative answers expected on orange notes: 'We can find acceleration from the gradient'; 'We can find total distance from area under the graph'; 'We can prove Runner A won because the area under Runner A's graph between km 25 and 35 is bigger.' Green notes might include: 'What if the graph is curved — not straight lines?'; 'Can we use this for the matatu on Nakuru highway — does it have smooth acceleration or steps?'. The teacher reads selected notes aloud and facilitates a 3-minute whole-class discussion to consolidate. The teacher explicitly links back to the phenomenon: 'We now have a mathematical answer to our driving question — not just a description, but a calculation that proves what happened at every point in the race.'</p> | <p>Distribute sticky notes. Say: 'You have 6 minutes in your groups. On the ORANGE note write what from today's lesson you can now use to answer our driving question about the Nairobi Marathon runners. On the GREEN note write a new question today's lesson raised. On the PURPLE note write what we still do not know.' [Circulate. Prompt groups whose orange notes are vague — e.g. 'we learnt about graphs' — with: 'Can you be more specific? Which calculation did you do today that helps explain why Runner A won?'] After 6 minutes, cold-call 4 groups to read their orange notes. Write a class summary on the board: 'From v-t graphs we can find: (1) acceleration = gradient; (2) distance = area under graph.' Ask: 'Before today, could we have proved mathematically why Runner B lost? What did we need that we did not have in Lessons 4 and 5?' [WAIT TIME: 12 seconds.] Affirm: 'Exactly — we needed the velocity-time graph and the area calculation. That is today's contribution to the driving question.'</p> | <p>Driving Question Board Protocol (Metacognitive Tracking): The three-colour sticky note system externalises the class's evolving understanding and makes thinking visible to all learners. The explicit distinction between 'what we can now answer' and 'what we still need' models scientific sensemaking — knowledge building is cumulative and iterative. Connecting the DQB update directly back to the marathon phenomenon reinforces that every lesson is a purposeful step toward resolving the anchoring puzzle, not an isolated skill exercise.</p> | <p>Teacher reads all orange sticky notes before the lesson ends. Adequate orange notes should reference: gradient, acceleration, area, and distance. Notes that only reference 'drawing graphs' indicate learners have not yet consolidated the interpretive skills and will need targeted re-engagement in Lesson 7. Green and purple notes are photographed (or transcribed) to inform lesson planning for Lessons 7 and 8.</p> |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b> <a href="#">Average velocity for constant acceleration</a></p> <p>Source: Khan Academy (English - US curriculum)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Learners individually update their cumulative motion model — a diagram they have been building since Lesson 1 to describe Runner A and Runner B's race. In Lessons 4–5 they added displacement-time graphs and speed calculations to the model. Today each learner adds: (i) a velocity-time graph</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Say: 'Get out your cumulative model from Lessons 1 to 5. Today you are going to add the velocity-time graph layer. I want you to sketch the velocity-time graph for each runner for the full race — label all phases. Then shade the km 25–35 section on each runner's graph and write the area</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Cumulative Model Revision (Progressive Model Building): Adding the velocity-time graph layer to an existing model — rather than starting fresh — makes the accumulation of knowledge physically visible and conceptually coherent. Learners can literally see the displacement-time graph</p>                                                                                                                                                                                                                                                                | <p>Teacher collects or photographs 6–8 models (stratified by ability group) to review before Lesson 7. Success criteria checked: (i) velocity-time graph present with velocity on y-axis and time on x-axis; (ii) at least two phases labelled (e.g. acceleration, constant velocity, deceleration);</p>                                                                                                                          |

|                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">Search ARES</a> for similar videos</p> <p><b>HTML:</b><br/> <a href="#">Displacement During Constant Acceleration (Flexbook)</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: CK-12<br/> <a href="#">Search ARES</a> for similar readings</p> | <p>sketch for each runner for the full race (labelling acceleration at the start, steady pace, water station slow-downs, and the final sprint); (ii) annotation arrows showing where gradient is steep (high acceleration) and shallow (slow acceleration); (iii) shaded regions on the km 25–35 segment of each runner's graph with a written label: 'Area = distance covered in km 25–35. Runner A's area &gt; Runner B's area. Therefore Runner A wins.' Learners write a 3-sentence model summary at the bottom: 'A velocity-time graph shows how velocity changes with time. The gradient of any segment tells us the acceleration or deceleration at that point. The area under the graph tells us the total distance travelled — and this is why Runner A, who had a higher velocity for longer between km 25 and 35, overtook Runner B and won the 2023 Nairobi Marathon.'</p> | <p>comparison. Finally, write your 3-sentence model summary. You have 8 minutes.' [Circulate. Look for: correct graph orientation (velocity on y-axis); correct labelling of phases; shaded region clearly marked; written comparison present.] With 2 minutes remaining: 'Swap your model with someone not in your group. Check: does their velocity-time graph correctly show Runner B starting faster? Does it show Runner A pulling ahead after the slowdown? Does their 3-sentence summary mention gradient AND area? Give one piece of specific written feedback.' [Allow 2 minutes for peer feedback.] Close: 'Hold up your model. Look at how much it has grown since Lesson 1 — we started with split times and a question. Now we have a mathematical proof. In Lesson 7 we will use what we know about velocity-time graphs to calculate relative speed — for example, when a matatu on the Nakuru highway overtakes a lorry.'</p> | <p>(from Lesson 5) sitting beside the velocity-time graph (from today), reinforcing that these are complementary descriptions of the same event. The peer-feedback protocol (checking for gradient label AND area label) operationalises the lesson's two key learning outcomes and gives every learner one last retrieval opportunity before the lesson ends.</p> | <p>(iii) km 25–35 shaded on both runners' graphs; (iv) written area comparison present and correct in direction (Runner A area &gt; Runner B area); (v) 3-sentence summary references both gradient (acceleration) and area (distance). Learners with fewer than 3 of 5 criteria met are flagged for a 5-minute catch-up at the start of Lesson 7.</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) GRADIENT CALCULATION — Were learners able to correctly identify the rise (change in velocity) and run (change in time) on the SGR graph? Did they apply correct units ( $\text{m/s}^2$  or  $\text{km/h per minute}$ )? If many learners wrote incorrect units, plan a 5-minute unit analysis warm-up at the start of Lesson 7. (2) AREA DECOMPOSITION — Which shape caused the most difficulty — the triangle, the rectangle, or the trapezium? If the trapezium caused widespread errors, prepare an additional worked example using a matatu velocity-time graph at the start of Lesson 7. (3) MARATHON CONNECTION — Did learners independently articulate, in Phase 4, that Runner A won because of greater area under the graph in the final 12 km? If this connection required significant teacher scaffolding, consider whether the phenomenon data (Runner A and Runner B velocity tables) needs to be made more concrete — e.g. providing numerical velocity values for each 5 km split rather than a qualitative description. (4) DQB ENGAGEMENT — Were green sticky notes (new questions) substantive and forward-looking (e.g. 'What if the graph is curved?')? Substantive green notes indicate high-level engagement and should be celebrated and used to frame Lesson 7. Superficial green notes ('I want to know more about graphs') indicate learners are not yet connecting the DQB to the driving question — revisit the DQB protocol briefly at the start of Lesson 7. (5) EQUITY MONITOR — Did all four members of each group contribute to the graph plotting and area calculation, or did one learner dominate? If group dynamics were unequal, restructure groups for Lesson 7 or assign specific roles (Plotter, Calculator, Recorder, Presenter) from the outset.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | Learners plotted and examined velocity-time graphs for two Kenyan real-world contexts: the SGR Madaraka Express journey out of Nairobi and a 400 m athletics race at Nakuru High School. They identified and labelled three motion phases — acceleration, constant velocity, and deceleration — by observing the shape and slope of each graph segment. They also observed a simplified velocity-time graph comparing Runner A and Runner B's motion between km 25 and 35 of the 2023 Nairobi Marathon.                                                                                                                                                                                                                          |
| <b>What did I learn?</b>                     | Learners learned that: (i) the gradient of a velocity-time graph equals acceleration — a steep positive gradient means rapid acceleration, a negative gradient means deceleration, and a zero gradient (horizontal line) means constant velocity; (ii) the area of the region between the velocity-time graph and the time axis equals the total distance (displacement) travelled; (iii) this area can be calculated by decomposing the region into rectangles and trapeziums, or by counting grid squares; (iv) comparing the areas under Runner A's and Runner B's velocity-time graphs for km 25–35 mathematically proves that Runner A covered more ground in that segment and therefore overtook Runner B to win the race. |
| <b>How does this explain the phenomenon?</b> | Learners explained, using their area calculations, why Runner B lost the 2023 Nairobi Marathon despite leading at the 30 km mark: Runner B's velocity dropped significantly between km 25 and 35 (smaller area under the graph = less distance covered per unit time), while Runner A maintained a higher velocity for longer (larger area = more distance covered). This resolved the central puzzle of the phenomenon and directly answered the key inquiry question 'How do we use velocity-time graphs?' Learners updated their cumulative models and the Driving Question Board to reflect this mathematical explanation.                                                                                                   |

## LESSON 7 (40 minutes): Relative Speed — Who Gets There First When Everyone Is Moving?

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners extend their understanding of motion from a single moving body to two bodies moving simultaneously, discovering that the speed one body appears to have depends entirely on the motion of the observer — and that this relative speed determines overtaking time, collision risk, and NTSA road-safety limits on Kenyan highways.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Knowledge</b>            | <ul style="list-style-type: none"> <li>– Define relative speed as the speed of one body as observed from another moving body.</li> <li>– State and apply the two rules: relative speed of bodies moving in the SAME direction = difference of their speeds; relative speed of bodies moving in OPPOSITE directions = sum of their speeds.</li> <li>– Solve problems involving time to overtake, time to meet, and distance apart using relative speed in Kenyan contexts (matatus on Nakuru highway, boats on Lake Victoria, runners on a track).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Skills</b>               | <ul style="list-style-type: none"> <li>– Calculate relative speed, time to meet, and time to overtake for two moving bodies given their individual speeds and initial separation.</li> <li>– Interpret real-world road-safety data (NTSA speed limits) using the mathematics of relative speed to justify why high closing speeds are dangerous.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Attitudes</b>            | <ul style="list-style-type: none"> <li>– Appreciate that mathematics is a tool for protecting human life — the same equations used to explain Runner B overtaking Runner A also explain why head-on collisions on Kenyan highways are catastrophic.</li> <li>– Show responsibility and integrity by recording calculations accurately and engaging honestly with road-safety implications.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Key Inquiry Question</b> | How do we use mathematics — specifically relative speed — to describe what one moving body experiences when another moving body approaches or passes it?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose in Storyline</b> | Lessons 1–6 built the complete mathematical toolkit for describing a SINGLE body in motion (distance, displacement, velocity, acceleration, displacement-time graphs, velocity-time graphs, and area under a graph). Lesson 7 now introduces the critical next layer: what happens when TWO bodies are moving at the same time? By using the matatu-overtaking supporting phenomenon — a matatu travelling at 90 km/h overtaking a bus travelling at 60 km/h on the Nakuru–Nairobi highway — learners discover that the speed each driver perceives the other to have is not their absolute speed but their RELATIVE speed. This bridges directly to the marathon phenomenon: Runner A overtook Runner B in the later part of the race (during the 30–35 km stretch), not because Runner A suddenly sped up, but because Runner B had slowed to about 17.6 km/h while Runner A held a steady 19 km/h, making the relative speed between them only about 1.4 km/h — the actual overtaking speed. Learners now have a complete quantitative explanation for every event in the 2023 Nairobi Marathon. |
| <b>Safety Notes</b>         | No specific physical hazards. When discussing road-safety contexts, ensure the conversation is sensitive — some learners may have been affected by road accidents. Frame discussion constructively around NTSA data and prevention, not graphic descriptions of crashes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

### B. LESSON OVERVIEW

Lesson 7 is the penultimate evidence-gathering lesson in the sequence and marks the conceptual shift from single-body motion to two-body relative motion. Having spent Lessons 1–6 building a complete mathematical toolkit — velocity, acceleration, displacement-time graphs, velocity-time graphs, and the area-under-graph method for distance — learners are now ready to ask a more sophisticated question: what does motion look like from the perspective of another moving body? The lesson is anchored by a vivid supporting phenomenon: a matatu travelling at 90 km/h attempting to overtake a Mombasa-bound bus travelling at 60 km/h on the Nakuru–Nairobi highway. Students must first predict whether the matatu driver can safely complete the overtake before an oncoming car arrives, then work through the mathematics of

relative speed to test their prediction.

The lesson introduces both cases of relative speed in a carefully sequenced order. Same-direction motion is treated first using the matatu-overtaking scenario: because both vehicles travel in the same direction, the matatu approaches the bus at only 30 km/h — its relative speed — not at 90 km/h. This explains why overtaking manoeuvres on Kenyan highways take far longer than inexperienced drivers expect, leading to the dangerous head-on collisions that NTSA statistics document every year. Opposite-direction motion is then explored using boats on Lake Victoria (a passenger ferry from Kisumu to Homa Bay meeting a cargo vessel travelling the opposite way) and runners on an athletics track. In the opposite-direction case the closing speed equals the SUM of the two speeds, which is why head-on crashes are so catastrophic.

Throughout the lesson learners connect back to the marathon anchoring phenomenon: Runner A's overtake of Runner B during the slow middle section (25–35 km) is now fully explained not just by the velocity-time graph shapes from Lesson 6, but by the precise relative speed between the two athletes at that moment. The Driving Question Board is updated to show that the driving question is now answerable in full, setting up Lesson 8 as the synthesis and consolidation lesson.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                    | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Finding Speed when Objects Travel in Opposite Directions</a><br><small>Source: CK-12</small><br> <a href="#">Search ARES for similar videos</a><br><br> <b>READING:</b><br><i>No closely matching reading in the ARES library for this activity.</i><br> <a href="#">Search ARES for readings</a><br><small>Search terms: linear motion relative speed</small> | Students are shown a short written scenario projected on the board (or read aloud): 'A matatu leaves Nakuru travelling towards Nairobi at 90 km/h. Exactly 0.5 km ahead on the same road, a bus is travelling in the same direction at 60 km/h. The matatu driver decides to overtake. An oncoming car is 3 km away, approaching at 100 km/h. Will the matatu complete the overtake safely before the oncoming car arrives?' Working individually in their exercise books, students write: (a) their gut-feel YES/NO answer, (b) one sentence explaining their reasoning, and (c) a numerical estimate of how long the overtake will take. Students are told not to worry about being right — this is about making their current thinking visible. | Display the scenario and say: 'Before we do any calculation, I want every single one of you to write your own answer — yes or no — and your reason. You have 2 minutes. No talking yet.' [WAIT 2 minutes, circulate and scan responses.] After 2 minutes: 'Keep your prediction visible — we will come back to it. Raise your hand if you said YES, the overtake is safe.' [Count hands, note number.] 'Raise your hand if you said NO.' [Count hands.] Cold-call 2 YES students and 2 NO students: 'Amina, walk us through your thinking in one sentence.' [WAIT 10 seconds after each response before calling the next student.] After four responses say: 'Interesting — we have disagreement in this room. That is exactly what good mathematics is for. Let us go find out who is right, and more importantly, WHY.' | Predict-before-observe (PBO) anchored to a Kenyan road scenario. By committing a prediction to paper before any instruction, every learner activates prior intuition about speed and creates a cognitive 'hook' — a genuine question they now want resolved. The public show-of-hands creates productive disagreement that motivates the observe phase. | Teacher scans written predictions for two things: (1) whether students are comparing speeds or only thinking about one vehicle's speed (diagnostic of whether they already have any intuition about relative motion), and (2) whether any student has spontaneously written 'difference' or 'closing speed' — indicating prior knowledge. Record the class split (YES vs NO) on the board to revisit after the Explain phase. |
| <b>Observe Phase</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Students work in pairs on a structured data-gathering activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Distribute/display the activity sheet and say: 'I am going to show you                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Structured Inquiry with Scaffolded Scenarios (graduated complexity).                                                                                                                                                                                                                                                                                    | Teacher listens for three specific misconceptions during circulation:                                                                                                                                                                                                                                                                                                                                                         |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p> <b>VIDEO:</b><br/> <a href="#">Finding Speed when Objects Travel in Opposite Directions</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/>  <a href="#">Search ARES for readings</a><br/> Search terms:<br/> linear motion<br/> relative speed</p> | <p>using three carefully sequenced worked scenarios printed on a single A4 activity sheet (or copied from the board into exercise books). SCENARIO 1 (Same direction — the matatu): Matatu speed = 90 km/h; Bus speed = 60 km/h; Gap = 0.5 km. Students are told: 'From the matatu driver's seat, the bus is NOT stationary — it is moving away at 60 km/h. The matatu gains on the bus at a rate of <math>(90 - 60) = 30</math> km/h. Use this closing speed to calculate how long it takes the matatu to cover the 0.5 km gap.' Students calculate <math>\text{time} = 0.5 \div 30 \text{ hours} = 1 \text{ minute}</math>. They then calculate how far the oncoming car travels in that same 1 minute at 100 km/h, and compare it to the 3 km gap. SCENARIO 2 (Opposite direction — Lake Victoria ferries): A passenger ferry leaves Kisumu at 25 km/h heading to Homa Bay. A cargo vessel leaves Homa Bay at 15 km/h heading to Kisumu. The ports are 60 km apart. Students use <math>\text{relative speed} = 25 + 15 = 40</math> km/h to find when and where the vessels meet. SCENARIO 3 (Runners — connecting to the marathon): During the slow middle section of the Nairobi Marathon (25–35 km), Runner A is travelling at 19 km/h and Runner B (who is 200 m ahead) has slowed to about 17.6 km/h. Students calculate the relative speed (about 1.4 km/h) and the time for Runner A to close the 200 m gap (about 8.6 minutes).</p> | <p>one key idea, and then you and your partner will apply it to all three scenarios.' Write on the board: 'SAME DIRECTION: Relative speed = Fast speed – Slow speed. OPPOSITE DIRECTIONS: Relative speed = Speed A + Speed B.' Say: 'This is the rule. Your job right now is not to memorise it — your job is to USE it on these three scenarios and see if the answers make sense to you.' [Allow 10 minutes of pair work. Circulate continuously.] After 5 minutes, pause the class: 'How many pairs have finished Scenario 1? Good. A quick check — cold-call: Juma, what closing speed did you get for the matatu and the bus?' [WAIT 10 seconds.] 'And Brian, does the matatu complete the overtake safely?' [WAIT 10 seconds.] After all three scenarios: 'Scenario 3 is the one I want you to think about most carefully. That is Runner A overtaking Runner B. Sound familiar?' [Pause 5 seconds — allow the connection to the marathon to land.] Cold-call one more pair: 'Fatuma, what is the relative speed of the two runners, and what does that number mean in plain language?'</p> | <p>The three scenarios are deliberately sequenced: matatu-overtaking (same direction, danger context), Lake Victoria ferries (opposite direction, distance-to-meet), and marathon runners (same direction, connecting back to the anchoring phenomenon). Each scenario requires students to identify which rule to apply before calculating, building procedural fluency while simultaneously demanding conceptual reasoning about direction. Pair work ensures every student articulates thinking aloud, not just the fastest workers.</p> | <p>(1) students adding speeds even in the same-direction case ('they are both fast, so add them') — redirect by asking 'from the matatu driver's window, is the bus getting CLOSER or FURTHER AWAY, and how FAST?'; (2) students forgetting to convert units (km/h to km/min or keeping consistent units) — check unit labels in working; (3) students who cannot connect Scenario 3 back to the marathon — prompt with 'Which runner is Runner A in our marathon story, and what was happening to Runner B's speed in the final 2 km according to the velocity-time graph we drew in Lesson 6?'</p> |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Interpreting direction of motion from</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>The class comes together for whole-group sensemaking. The teacher leads a structured discussion that formalises the two relative-speed rules, connects</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Begin by returning to the board prediction tally: 'At the start of the lesson, [X] of you said the overtake was safe and [Y] said it was not. Let us now use the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Claim–Evidence–Reasoning (CER) structured discussion. Teacher explicitly models the three-step CER pattern for each problem: CLAIM (state the</p>                                                                                                                                                                                                                                                                                                                                                                                        | <p>Teacher collects two observable indicators: (1) During the piki-piki problem, can students independently identify that same-direction subtraction applies</p>                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">velocity-time graph</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar videos</p> <p> <b>READING:</b><br/> <i>No closely matching reading in the ARES library for this activity.</i><br/> <a href="#">Search ARES</a> for readings<br/> Search terms: linear motion relative speed</p> | <p>them to the NTSA road-safety context, and then returns to the original prediction to resolve it. Students annotate their activity sheets with formal definitions and mark whether their original predictions were correct. The class then works through two extension problems: (a) A piki-piki (motorcycle) travelling at 70 km/h is chased by a police car travelling at 110 km/h. They start 1.2 km apart. How long before the police car catches the motorcycle? (b) Two matatus leave Nairobi and Mombasa simultaneously, 480 km apart, travelling towards each other at 80 km/h and 100 km/h respectively. When and where do they meet? Students solve these in their exercise books, then a volunteer writes the solution on the board for class verification.</p> | <p>mathematics to settle this.' Work through Scenario 1 on the board step by step, narrating each line: 'Step 1: Identify direction — same direction, so we SUBTRACT. Step 2: Relative speed = <math>90 - 60 = 30</math> km/h. Step 3: Time to close 0.5 km gap at 30 km/h = <math>0.5/30</math> hours = 1 minute. Step 4: In 1 minute, the oncoming car travels <math>100 \times (1/60) = 1.67</math> km. Step 5: The oncoming car started 3 km away and has covered 1.67 km — it is now only 1.33 km away. So what is your verdict?' [WAIT 15 seconds, cold-call 2 students.] Then say: 'Now I want you to tell me WHY head-on collisions are so much more deadly than rear-end collisions. Use the words RELATIVE SPEED in your answer.' [WAIT 15 seconds.] Cold-call 3 students. Formalise: 'Head-on collision relative speed = <math>80 + 80 = 160</math> km/h. Rear-end: <math>90 - 60 = 30</math> km/h. The force of impact depends on this relative speed. That is why NTSA sets speed limits — it is not arbitrary, it is mathematics protecting lives.' For the extension problems, say: 'In your books — 3 minutes. Go.' [Circulate.] After 3 minutes: 'Wanjiku, come and write your solution for the police car problem. Everyone else, check your own work as she writes.'</p> | <p>answer), EVIDENCE (show the calculation), REASONING (explain why the rule applies — same direction subtract, opposite directions add). Students are required to use this same structure when explaining their extension-problem solutions. This is the same scientific-reasoning structure embedded throughout the ARES sequence and reinforces the habit of justifying answers, not just stating them.</p> | <p>without being prompted? (2) In the Nairobi–Mombasa matatu problem, can students correctly set up distance = relative speed <math>\times</math> time AND then calculate where the meeting point is (not just when)? Students who can do both are at 'Meets Expectations'; students who also correctly interpret the road-safety implication in their own words are at 'Exceeds Expectations'. Students who confuse the two rules need targeted same-day intervention using the direction-of-motion visual (arrows on number line) drawn on a mini-whiteboard.</p> |
| <p><b>Driving Question Board (DQB) Creation</b><br/> <b>VIDEO:</b><br/> <a href="#">Interpreting direction of motion from position-time graph</a><br/> <i>Related topic: not</i></p>                                                                                                                                                                                                                                   | <p>Students return to the class Driving Question Board displayed at the front of the room. They review all the sticky notes and annotations accumulated across Lessons 1–6. Working in their table groups (3–4 students), each group receives two sticky notes: one GREEN ('We can now answer this') and one ORANGE ('New question this lesson raised').</p>                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Say: 'Take 3 minutes in your groups. Look at our DQB. Which questions can we NOW answer that we could not answer before Lesson 7? Write your answer on the GREEN note. Then think: did today's lesson raise a NEW question for you? Write it on the ORANGE note.' [WAIT 3 minutes. Circulate and prompt quiet groups: 'What about Runner A overtaking</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Driving Question Board Annotation (Progressive Refinement). By colour-coding sticky notes as GREEN (resolved) and ORANGE (new questions), students experience the scientific process authentically: answering one question always generates new ones. This also provides the teacher with a real-time class-level formative snapshot of which</p>                                                           | <p>Teacher photographs the DQB after the lesson. Key indicator: can at least 70% of groups correctly identify the Runner A / Runner B overtaking question as now answerable using relative speed? If groups are still posting this as an ORANGE (unresolved) note, it signals that the connection between Scenarios 1/3 and the marathon phenomenon was not</p>                                                                                                                                                                                                     |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>an exact match for this lesson.</i><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/><i>No closely matching reading in the ARES library for this activity.</i><br/><a href="#">Search ARES for readings</a><br/>Search terms:<br/>linear motion<br/>relative speed</p>                                                                                                                            | <p>Groups discuss which existing DQB questions can now be fully answered using relative speed (e.g. 'How did Runner A overtake Runner B if they were both running at high speed?'), and write any new questions about real-world applications (e.g. 'Does the NTSA use relative speed equations to set speed limits?' or 'What happens to relative speed when one car brakes suddenly?'). Groups post their sticky notes and one spokesperson reads them aloud to the class.</p>                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Runner B — could you explain that fully before today? Can you now?'] After groups post notes, go to the board and point to the driving question: 'How can we use mathematics — calculations and graphs — to FULLY describe and explain the motion of ANY moving body?' Cold-call one student: 'Otieno — after seven lessons, can we now answer this question? What is still missing?' [WAIT 15 seconds.] Acknowledge any remaining gaps (e.g. 'We have not yet done a full problem combining graphs AND relative speed for the marathon scenario — that is Lesson 8.') Say: 'Our DQB shows us we are almost there. One lesson to go.'</p>                                                                                                                                                                                                                                                          | <p>concepts have genuinely landed versus which remain unclear, shaping the design of Lesson 8.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>made explicit enough and must be revisited in the opening of Lesson 8.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p><b>Model Building Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Instantaneous speed and velocity</a><br/><i>Related topic: not an exact match for this lesson.</i><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>READING:</b><br/><i>No closely matching reading in the ARES library for this activity.</i><br/><a href="#">Search ARES for readings</a><br/>Search terms:<br/>linear motion<br/>relative speed</p> | <p>Each student returns to their cumulative 'Motion Model' — the annotated diagram they have been building since Lesson 1 that represents the complete story of the 2023 Nairobi Marathon using every mathematical tool learned so far. In Lesson 7, students add a NEW layer to their model: a two-body diagram showing the overtake during the 30–35 km stretch. They draw two parallel horizontal arrows (representing Runner A and Runner B moving in the same direction), label the arrows with the velocities from Lesson 6's velocity-time graph (Runner A: 19 km/h, Runner B: 17.6 km/h), write the relative speed calculation beside the diagram (<math>19 - 17.6 = 1.4</math> km/h), and annotate the 200 m gap with the time calculation showing when Runner A catches Runner B. Below the diagram they write a one-sentence 'mathematical story' of the overtake in plain English, for</p> | <p>Say: 'Open your Motion Models. Today we add the final piece of the mathematical story of the marathon — the moment Runner A overtook Runner B. I want you to draw a two-body diagram. Use the velocity values from your velocity-time graph in Lesson 6.' [Display the relevant section of the Lesson 6 v-t graph on the board as a reference.] 'You have 4 minutes. Draw, calculate, annotate, and write your one-sentence story.' [WAIT 4 minutes, circulate and look at models.] After 4 minutes, cold-call one student to read their one-sentence story aloud. Ask the class: 'Is this a complete mathematical explanation of why Runner A won? What did we use from Lesson 1? Lesson 2? Lesson 3 and 4? Lesson 5 and 6? And now Lesson 7?' [WAIT 10 seconds between each lesson reference — students call out or raise hands.] Conclude: 'Your model now contains distance, displacement,</p> | <p>Cumulative Model Revision (Progressive Modelling). The running Motor Model that students have built across all seven lessons functions as a personalised scientific model in the NGSS sense: it is revised and extended each lesson as new evidence is gathered and new concepts are learned. Adding the two-body relative-speed layer in Lesson 7 makes the model 'complete' — every event in the marathon phenomenon now has a mathematical representation in the model. The one-sentence 'mathematical story' annotation develops scientific communication skills and forces students to translate symbolic mathematics into coherent English prose.</p> | <p>Teacher collects (or photographs) 4–5 models from different ability groups. Check for three things: (1) correct identification of which runner is faster and correct application of the subtraction rule (not addition); (2) correct unit handling in the time calculation (converting km/h and km to find time in hours, then converting to minutes); (3) the one-sentence story accurately uses the phrase 'relative speed' or 'from Runner B's perspective' — indicating conceptual understanding, not just procedural calculation. Students whose models show correct calculation but incorrect prose (or no prose) are directed to the 'Tell Me the Story' prompt card: 'If Runner B could feel Runner A approaching, what speed would Runner A seem to be running at?'</p> |

|  |                                                                                                                                                                                  |                                                                                                                                                                                                                                       |  |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  | example: 'From Runner B's perspective, Runner A appeared to be approaching at only 1.4 km/h — but that was enough to close the 200 m gap in about 8.6 minutes and win the race.' | velocity, acceleration, a displacement-time graph, a velocity-time graph, area under the graph for distance, AND relative speed. That is the complete mathematical toolkit for linear motion. In Lesson 8, we bring it all together.' |  |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

### D. TEACHER REFLECTION

1. When students worked through Scenario 1 (the matatu overtaking), were they able to independently identify that same-direction motion requires SUBTRACTION before I stated the rule, or did they wait for me to give the formula? What does this tell me about their readiness for Lesson 8's synthesis task?
2. Were the three scenarios sufficiently differentiated in difficulty and context? Did the Lake Victoria ferry scenario (Scenario 2) cause more confusion than the road scenarios — and if so, is it because the context was unfamiliar, or because the 'distance to meeting point' calculation added an extra step beyond what students were ready for?
3. When I asked students to explain WHY head-on collisions are more dangerous than rear-end collisions using the words 'relative speed', how many students gave a calculation-based answer versus a vague intuitive answer? This reveals whether the concept is conceptually understood or only procedurally applied.
4. Looking at the Driving Question Board after the lesson: how many GREEN notes correctly identified the Runner A / Runner B overtake as now explainable? Were there any ORANGE notes that revealed genuine new mathematical questions I had not anticipated — and if so, can I address them in Lesson 8?
5. In the Model Building phase, did students connect their new two-body diagram back to the velocity-time graph values from Lesson 6 without prompting? If students needed to be reminded to use Lesson 6 values, I need to build a stronger explicit cross-lesson reference at the START of Lesson 8 before the final synthesis task begins.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | We observed (through calculation) that a matatu travelling at 90 km/h approaches a bus travelling in the same direction at 60 km/h at a closing speed of only 30 km/h — meaning the overtake takes much longer than the driver expects. We also observed that two Lake Victoria ferries travelling towards each other close the gap between them at 40 km/h (the SUM of their speeds), and that Runner A appeared to approach Runner B at just 1.4 km/h when overtaking — the DIFFERENCE of their velocities.                                                                                                                                                                                                                                                              |
| <b>What did I learn?</b>                     | We learned that RELATIVE SPEED is the speed of one body as observed from another moving body. When two bodies move in the SAME direction, relative speed = faster speed – slower speed (they approach each other slowly). When two bodies move in OPPOSITE directions, relative speed = speed A + speed B (they approach each other rapidly). We can use relative speed to calculate the time for two bodies to meet or for one to overtake the other, using the formula: time = gap distance ÷ relative speed.                                                                                                                                                                                                                                                            |
| <b>How does this explain the phenomenon?</b> | This explains a crucial part of the marathon phenomenon: Runner A did not need to run faster than Runner B's earlier pace to overtake him — Runner A only needed to run faster than Runner B's CURRENT speed. Because Runner B had slowed to about 17.6 km/h and Runner A was running at about 19 km/h, the relative speed between them was about 1.4 km/h. At that closing rate, Runner A covered the 200 m gap in approximately 8.6 minutes. The same mathematics also explains the NTSA road-safety context: a head-on collision between two vehicles both doing 80 km/h has a relative speed of 160 km/h — four times more dangerous than a rear-end collision between a 90 km/h matatu and a 60 km/h bus. Mathematics does not just describe motion — it saves lives. |



## LESSON 8 (80 minutes): Consolidation, Model Building & Final Explanation: The Full Story of the Nairobi Marathon

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | To consolidate all learning in Sub-Strand 2.9 by producing a complete mathematical explanation of the Nairobi Marathon phenomenon — integrating calculations of speed, velocity and acceleration with displacement-time and velocity-time graphs — and to demonstrate this understanding through group model presentations, DQB closure, and a class written assessment.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Knowledge</b>            | Learners demonstrate knowledge of: distance, displacement, speed, velocity, acceleration and relative speed; the construction and interpretation of displacement-time graphs (gradient = velocity, shape = story of motion); the construction and interpretation of velocity-time graphs (gradient = acceleration, area = displacement); and the distinction between average speed and instantaneous velocity across a 42 km race.                                                                                                                                                                                                                                                                                                                                                               |
| <b>Skills</b>               | Learners accurately calculate average speed, instantaneous velocity, and acceleration from split-time tables; plot and annotate displacement-time and velocity-time graphs for two runners on the same axes; interpret graph shapes (steepness, curvature, horizontal segments) to narrate motion; compare models built in Lesson 1 with final models to show conceptual growth; communicate mathematical reasoning in written and oral form.                                                                                                                                                                                                                                                                                                                                                    |
| <b>Attitudes</b>            | Learners appreciate the power of mathematics to reveal stories hidden in data; value integrity in accurate data recording and graph plotting; demonstrate unity by collaborating in group model presentations; show responsibility by completing the written assessment independently and honestly; and appreciate how linear motion mathematics connects to careers and everyday Kenyan life.                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Key Inquiry Question</b> | 1. What is speed? 2. How do we use displacement-time graphs in real life? 3. How do we use velocity-time graphs?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Purpose in Storyline</b> | This is the final lesson of the storyline. Learners return to the original Nairobi Marathon split-time data that launched the driving question in Lesson 1. Armed with all the mathematical tools built across Lessons 1–7, they now produce a complete, evidence-based explanation of why Runner A won despite Runner B leading at the 30 km mark — using precise calculations, dual-runner graphs, and written interpretation of the critical sprint finish. Groups present their final models, the class closes every question on the Driving Question Board, and a written assessment consolidates all sub-strand outcomes. The lesson ends with learners reflecting on careers that use linear motion mathematics, anchoring the mathematics in the real world of Kenyan work and industry. |
| <b>Safety Notes</b>         | No physical hazards in this lesson. Learners should handle graph paper, rulers and calculators with care. Ensure equitable group participation — monitor that quieter learners are not sidelined during group presentations. During the written assessment, enforce a calm, respectful environment so all learners can demonstrate their individual best.                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### B. LESSON OVERVIEW

Lesson 8 is the culminating lesson of Sub-Strand 2.9: Linear Motion. The lesson opens by returning dramatically to the 2023 Nairobi Marathon video clip and printed split-time table that launched the unit — asking learners: ‘You now have all the tools. Can you finally explain exactly what happened at every kilometre of that race?’ In the Predict phase, learners compare their current mental model to their Lesson 1 prediction and articulate what they now know. In the Observe phase, groups receive the full Runner A and Runner B split-time data and produce a complete mathematical portfolio: average speed calculations, segment-by-segment velocity, acceleration at the sprint finish, a dual displacement-time graph, and a dual velocity-time graph plotted on the same axes. In the Explain phase, each group presents their model to the class — teacher facilitates cross-group critique and connects every interpretation back to the phenomenon. The DQB Creation phase closes all three KICD key inquiry questions with class-agreed, evidence-based answers drawn directly from the marathon data. The Model Building phase places learners’ Lesson 1 sketch beside their final graph, making conceptual growth visible. A 20-minute class written assessment covers all sub-strand SLOs. The lesson closes with a structured career

reflection: learners brainstorm and discuss Kenyan careers — civil engineer designing Thika Superhighway, NTSA road safety analyst, track and field coach at Kipchoge Keino Stadium, Kenya Railways planner — that depend on the mathematics of linear motion.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                    | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Position vs. time graphs</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Instantaneous velocity and instantaneous speed from graphs</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners view the opening 90 seconds of the Nairobi Marathon video clip again — the same clip from Lesson 1. They then receive back their original Lesson 1 prediction sheet (kept by the teacher) showing what they wrote about Runner A and Runner B before instruction began. Working individually for 3 minutes, learners write a 'Then vs. Now' comparison: (1) What did I think was happening in Lesson 1? (2) What do I know now that I did not know then? (3) In one sentence, what is my complete explanation of why Runner A won? Learners then share their 'Then vs. Now' with a shoulder partner for 2 minutes before two or three are cold-called to share with the class. This activates prior knowledge, surfaces conceptual growth, and creates emotional investment in the final explanation task. | Teacher replays the marathon clip on the projector and says: 'This is exactly where we started eight lessons ago. You had a puzzle — same distance, different stories, Runner B was winning at 30 km but Runner A crossed the finish line first. Look at what you wrote in Lesson 1.' Distribute Lesson 1 prediction sheets. Set a 3-minute silent individual writing timer. After 3 minutes, say: 'Turn to your shoulder partner. Share your Then vs. Now. You have 2 minutes.' WAIT for full 2 minutes. After pairs share, cold-call exactly 3 learners using a name stick: 'Tell us in one sentence — why did Runner A win?' After each response, probe with: 'How do you know that mathematically?' Record the three responses on the board under the heading 'Our Current Best Explanation.' Do NOT confirm or correct yet — hold the tension for the Observe phase. Estimated time: 10 minutes. | Predict-Observe-Explain (POE) Activation with 'Then vs. Now' Comparison. By physically holding their Lesson 1 prediction and comparing it to current understanding, learners make their own conceptual growth visible and concrete. This metacognitive move (recognising 'I used to think... now I think...') is a powerful sensemaking strategy that motivates the final explanation task and frames the entire lesson as the resolution of a journey. | Teacher listens to the three cold-called responses and checks: (a) Does the learner reference velocity rather than just speed? (b) Does the learner mention the sprint finish or changing rate? (c) Does the learner use the word 'graph' or 'gradient'? Responses that remain at 'Runner A was faster' (without qualification) signal that the learner has not yet distinguished average speed from instantaneous velocity — the teacher notes these learners for targeted support during the Observe phase group work. |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Position vs. time graphs</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b>                                                                                                                                                                                                                                                                 | Learners work in pre-assigned groups of four. Each group receives: (1) the complete Runner A and Runner B split-time data table (time in minutes recorded at 0, 5, 10, 15, 20, 25, 30, 35, 40, and 42 km marks); (2) a large sheet of graph paper (A3); (3) a structured portfolio task sheet with four required outputs. Task 1 — Calculations: Compute the                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Distribute materials. Read the task aloud once. Say: 'You have 25 minutes. Every member of the group must be able to explain any part of this portfolio — I will cold-call individuals, not groups, when you present.' Set a visible timer. Circulate and use targeted prompts: For Task 1 — 'What does a steeper gradient on a displacement-time graph tell you                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mathematics Portfolio as Evidence Synthesis. Rather than a single task, the portfolio forces learners to move iteratively between numerical data (Task 1), graphical representations (Tasks 2 and 3), and verbal explanation (Task 4). This multi-representational approach — numbers → graphs → words — mirrors the practice of professional mathematicians and                                                                                        | Teacher uses a mental checklist during circulation: (a) Are displacement-time graphs plotted with correctly scaled axes and labelled units? (b) Are segment velocities correctly calculated (change in displacement ÷ change in time)? (c) Does the velocity-time graph show Runner B's velocity dropping in the 25–35 km segment and spiking for the sprint? (d)                                                                                                                                                        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">Average velocity and average speed from graphs</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                   | <p>average speed for the whole race for both runners (km/h). Then compute the average velocity for each 5 km segment. Identify the segment where Runner B decelerates most sharply (between 25–35 km) and compute the acceleration. Task 2 — Displacement-Time Graph: Plot both runners on the same axes (time on x-axis in minutes, displacement on y-axis in km). Use different colours. Annotate: the point where Runner B is ahead (at 30 km), the sprint finish zone (40–42 km), and where the lines cross (Runner A overtakes). Task 3 — Velocity-Time Graph: Using segment velocities computed in Task 1, construct a velocity-time graph for both runners on the same axes. Annotate: Runner B's deceleration zone, Runner A's consistent gradient, and the sprint finish. Task 4 — Written Explanation: In 5–7 sentences, explain in mathematical language why Runner A won, referencing gradient, velocity, and acceleration. Groups have 25 minutes for all four tasks. The teacher circulates actively.</p> | <p>about velocity? Show me where that is in your data.' WAIT 12 seconds after asking. For Task 2 — 'Use a ruler. The gradient of each segment IS the velocity. Check: does your graph's steepness match the velocities you calculated?' For Task 3 — 'Where is Runner B's velocity going down? What does that look like on a velocity-time graph?' For Task 4 — 'Avoid saying faster or slower. Use the words gradient, velocity, and acceleration.' If a group finishes early, challenge: 'Calculate the area under Runner A's velocity-time graph for the first 20 km. What does that area represent?' Spend no more than 3 minutes with any single group. Call a 5-minute warning. Estimated time: 27 minutes (2 min setup + 25 min group work).</p> | <p>scientists and builds robust understanding by requiring learners to translate across modes. Each translation acts as a self-check: if the graph does not match the calculation, something is wrong and must be reconciled.</p>                                                                                                                                                                                                                                                                     | <p>Does the written explanation include the word 'gradient' and a reference to acceleration? Groups that cannot connect the gradient of the displacement-time graph to velocity receive a targeted prompt card: 'Gradient = rise ÷ run = <math>\Delta\text{displacement} \div \Delta\text{time} = \text{velocity}</math>. Apply this to your graph now.' Teacher identifies the group with the most precise Task 4 explanation to present first.</p>                                                                |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b> <a href="#">Acceleration vs. time graphs</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b> <a href="#">Instantaneous velocity and</a></p> | <p>Each group has 3 minutes to present their portfolio to the class. One member holds up the A3 graph; another reads the written explanation; a third fields questions. Groups post their A3 graphs on the classroom wall (gallery style) after presenting. After all groups present, learners do a 3-minute silent 'gallery walk' — they move to look at the other groups' graphs on the wall and place a sticky dot next to the feature they find most clearly explained. The class then</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Announce: 'Each group has exactly 3 minutes. I will time you. After your presentation, post your graph on the wall.' Use a stopwatch strictly. After each presentation, cold-call one individual from the audience (not the presenting group): 'In one sentence, what is the most important thing that group's graph showed you?' WAIT 10 seconds. After all presentations and the gallery walk, stand at the board and say: 'Let's build one final class explanation together. I need three</p>                                                                                                                                                                                                                                                     | <p>Gallery Walk with Sticky Dot Critique and Class Consensus Construction. The gallery walk externalises every group's thinking simultaneously, allowing learners to compare representations side-by-side. Peer critique (sticky dots on the clearest features) develops evaluative reasoning. Constructing the final class explanation verbally and then in writing on the board models the scientific practice of argumentation from evidence (NGSS SEP 7: Engaging in Argument from Evidence),</p> | <p>Teacher listens for precision in the three cold-called audience responses after each presentation. Key indicators: (a) Learner uses 'gradient' rather than 'steepness' when describing the graph. (b) Learner references a specific km segment rather than 'the middle of the race.' (c) Learner connects the velocity-time graph area to displacement. Vague responses ('Runner B slowed down') are probed: 'How does the graph show that? Point to the specific feature.' The class consensus statement on</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><a href="#">instantaneous speed from graphs</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                      | <p>reconvenes and the teacher facilitates a 5-minute whole-class discussion using the gallery walk observations to identify: (1) the single clearest displacement-time graph, (2) the single clearest velocity-time graph, and (3) the single most precise written explanation. These three exemplars are used to build the class consensus final explanation on the board.</p>                                                                                                                                                                                                                 | <p>volunteers — one to describe the displacement-time graph, one to describe the velocity-time graph, and one to explain the sprint finish in terms of acceleration.' Write their contributions verbatim, then refine together as a class. Final board statement should read approximately: 'Runner A maintained a nearly constant velocity of approximately X km/h throughout the race. Runner B's velocity decreased significantly between km 25 and km 35, shown by the shallower gradient on the displacement-time graph and the negative slope on the velocity-time graph. In the final 2 km sprint, Runner B's velocity increased (positive acceleration) but it was insufficient to overcome the time deficit. Runner A's total time was lower because of consistent velocity, not a faster sprint.' Estimated time: 18 minutes (3 groups × 3 min + 1 min transition each = ~12 min; gallery walk = 3 min; discussion = 5 min — adjust group count as needed).</p> | <p>grounding the argument in the marathon data.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>the board becomes a reference for the assessment phase.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b> <a href="#">Deriving displacement as a function of time, acceleration, and initial velocity</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> | <p>The full Driving Question Board — which has been built up across all 8 lessons — is displayed at the front of the class. The board shows the original driving question, all three KICD key inquiry questions, and every sticky-note question added by learners across the unit. Working as a whole class, learners now formally close each question. The teacher reads each sticky note aloud. For the three KICD key inquiry questions, the class agrees on a final answer — drawn from the marathon evidence — and the teacher writes it in a coloured marker on a card placed next to</p> | <p>Stand at the DQB and say: 'Eight lessons ago, you asked how we can use mathematics to explain the motion of any moving body. Today you are going to answer that question with evidence.' Read the first KICD question: 'What is speed?' Ask: 'Someone give me our final, mathematically precise answer — using the marathon data.' WAIT 12 seconds. Cold-call. Write the answer on a green card and pin it. Repeat for questions 2 and 3. For sticky-note questions, move quickly — read each, ask for a show of hands: 'Answered — yes or no?' Mark accordingly. For the driving question, say:</p>                                                                                                                                                                                                                                                                                                                                                                   | <p>Driving Question Board Closure as Metacognitive Summary. Revisiting every question on the DQB and formally marking it 'answered' or 'for further study' gives learners a concrete, visual record of their intellectual journey. This practice mirrors the scientific community's norms of knowledge-building and revision. It also validates learner curiosity — every sticky note is taken seriously — and models that unanswered questions are not failures but invitations to further inquiry (NGSS SEP 1: Asking Questions and Defining Problems).</p> | <p>Teacher notes which questions learners mark as 'unanswered' even though they have been addressed in prior lessons — this reveals residual misconceptions. Particular attention to: (a) 'Is velocity the same as speed?' — if learners are uncertain, the teacher clarifies verbally before the assessment: 'Velocity includes direction; speed does not. A runner going back to collect a water bottle has negative velocity on our displacement-time graph.' (b) 'What does the area under a velocity-time graph mean?' — if uncertain, teacher points to the graph on the wall and traces the</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p> <b>EXERCISE:</b><br/> <a href="#">Average velocity and average speed from graphs</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>that question. For any unanswered sticky-note questions ('What happens when two runners run in opposite directions?' / 'What does deceleration look like on a velocity-time graph?'), learners vote: (a) 'We answered this — where?' or (b) 'This is a question for further study.' Unanswered questions are circled in a different colour. The driving question itself is answered last, with the full class reciting the consensus explanation built in Phase 3.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>'Everyone, together, say our final answer.' Lead the class in a choral recitation of the consensus explanation. Then step back and say: 'Look at this board. Every question you ever had about linear motion in this unit is answered or deliberately saved for the next level. That is what mathematicians do.' Estimated time: 8 minutes.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>area with a finger.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Model Building Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Average velocity for constant acceleration</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Instantaneous velocity and instantaneous speed from graphs</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Part A — Model Comparison (5 minutes): Learners place their Lesson 1 rough sketch (of what they thought Runner A and Runner B's motion looked like) beside their final, fully labelled, dual-runner displacement-time graph from today. On a reflection slip, they complete two sentences: 'In Lesson 1, my model showed... because I thought...' and 'My final model now shows... because the mathematics tells us...' These slips are collected as a formative artefact. Part B — Written Assessment (20 minutes): Learners complete an individual written assessment covering all sub-strand SLOs. Questions include: (Q1) Define velocity and distinguish it from speed, giving an example from the Nairobi Marathon. (Q2) A matatu travels from Nakuru to Naivasha (90 km) in 1.5 hours, stops for 10 minutes, then travels 60 km to Nairobi in 45 minutes. Draw the displacement-time graph and calculate the average speed for the entire journey. (Q3) From the velocity-time graph provided, calculate the acceleration in the first 10 minutes and find the total displacement for</p> | <p>Distribute Lesson 1 sketches and reflection slips. Say: 'You have 5 minutes. Place your Lesson 1 sketch and your final graph side by side. Complete the two sentences on your reflection slip. Be honest — growth in understanding is what we celebrate here.' Collect slips after 5 minutes. Distribute assessment papers. Say clearly: 'This is individual work. No group discussion. You have exactly 20 minutes. Use the class consensus explanation on the board as a reminder of your mathematical vocabulary — but all working must be your own.' Set a timer. Circulate silently, noting learners who appear stuck on Q2 (common error: forgetting to include the stationary period on the graph). Do not give answers but may prompt: 'Read the question again. How many stages does this journey have?' With 5 minutes remaining, give a verbal warning. Collect all papers. Open the career reflection chart and ask: 'Shout out a Kenyan career that uses what you learned.' Scribe rapidly for 3 minutes. Close with: 'The Nairobi Marathon runners and the matatu drivers on the Nakuru highway do not know</p> | <p>Model Comparison as Visible Learning Growth (Lesson 1 vs. Final Model). Placing the naive Lesson 1 sketch beside the final mathematically rigorous dual-runner graph makes conceptual change physically visible to each learner. This is a powerful metacognitive sensemaking strategy: learners see not just what they now know, but how their thinking changed and why — because the evidence from the marathon data demanded it. The written assessment then provides individual accountability for the knowledge, skills and attitudes developed across the unit. The career reflection anchors the mathematics in authentic Kenyan professional contexts, fulfilling the KICD value of connecting education to the world of work (NGSS SEP 8: Obtaining, Evaluating and Communicating Information).</p> | <p>Reflection slips are collected and reviewed: teacher checks whether the 'final model' sentence includes at least one mathematical term (gradient, velocity, acceleration, displacement). Written assessment Q2 and Q3 are the primary summative indicators for graph-drawing and graph-interpretation SLOs. Q4 (relative speed) tests the final sub-strand outcome. Q5 (graph interpretation in the marathon context) closes the loop on the phenomenon. Rubric levels (Exceeds / Meets / Approaches / Below Expectations) align to the KICD suggested assessment rubric for Linear Motion: Ability to determine displacement, speed, velocity, acceleration; Ability to draw and interpret graphs of linear motion; Ability to solve problems involving relative speed.</p> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                        |  |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  | <p>the 40-minute journey. (Q4) Two matatus leave Kisumu and Eldoret simultaneously, travelling toward each other at 80 km/h and 60 km/h respectively. The distance between the two towns is 280 km. At what time do they meet, and how far from Kisumu? (Q5) Interpret the shape of Runner B's displacement-time graph between km 25 and km 35: what does the shallower gradient tell you about Runner B's motion? Part C — Career Reflection (5 minutes, post-assessment): Learners brainstorm on a class chart: 'Which Kenyan careers use linear motion mathematics every day?' Teacher scribes responses — NTSA road safety analyst, Kenya Railways planner, civil engineer on the Nairobi Expressway, athletics coach at Kipchoge Keino Stadium in Eldoret, SGR locomotive driver, Nairobi Marathon race director. Teacher closes: 'Every career on this list needs exactly what you built in this unit — the ability to read, draw, and explain motion using numbers and graphs.'</p> | <p>they are living inside a mathematics lesson — but you do. That is the power of what you have learned.' Estimated time: 5 min model comparison + 20 min assessment + 5 min careers = 30 minutes.</p> |  |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

## D. TEACHER REFLECTION

After Lesson 8, reflect on the following questions before closing the unit: (1) Phenomenon closure — Did every learner produce a written sentence that explains why Runner A won using at least two mathematical terms (e.g., velocity and gradient)? If not, which learners need individual follow-up? (2) Graph accuracy — Were the majority of dual-runner displacement-time graphs correctly scaled, labelled with units, and annotated with the overtaking point and sprint finish zone? Common errors to note for future units: forgetting units on axes, drawing straight lines where curves are appropriate (e.g., the sprint finish), and confusing the x-axis label (time) with the y-axis label (displacement). (3) DQB effectiveness — Did the sticky-note questions evolve across the 8 lessons in a way that reflected genuine conceptual growth? Were there persistent questions (e.g., 'What is the difference between speed and velocity?') that suggest the unit needs earlier, more explicit treatment? (4) Assessment data — Review Q2 (displacement-time graph with a stop) and Q4 (relative speed) as key diagnostic items. If more than 30% of learners score Below Expectations on Q4, consider adding one consolidation lesson on relative speed in the next iteration of this unit. (5) Career relevance — Did learners connect linear motion to Kenyan careers authentically, or did responses remain generic? In future, invite a guest speaker — e.g., an NTSA road safety officer or a Kenya Railways engineer — to speak to the class during this lesson's career reflection segment, making the connection between school mathematics and professional practice vivid and personal. (6) Storyline integrity — Did the return to the Nairobi Marathon phenomenon in Lesson 8 feel like a satisfying resolution of the question posed in Lesson 1, or did it feel procedural? The emotional payoff of 'now I can explain this' is pedagogically essential — if it felt flat, consider adding a brief moment where the original video is paused mid-clip and a learner narrates what is mathematically happening on screen in real time.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | Learners re-watched the 2023 Nairobi Marathon video clip showing elite Kenyan runners — including Runner A and Runner B — with different pacing strategies. They examined the printed split-time table showing each runner's time at every 5 km mark from 0 to 42 km, observing that Runner B led at 30 km but Runner A finished first. Groups plotted dual displacement-time and velocity-time graphs using the split-time data, and posted their A3 graphs on the classroom wall for a gallery walk comparison.                                                                                                                                                                |
| <b>What did I learn?</b>                     | Learners produced a complete mathematical explanation of the marathon race, calculating average speed, segment velocity, and acceleration; drawing and interpreting displacement-time graphs (where gradient equals velocity) and velocity-time graphs (where gradient equals acceleration and area equals displacement); distinguishing between average speed and instantaneous velocity; and applying relative speed concepts. They closed all three KICD key inquiry questions on the Driving Question Board with evidence-based answers drawn directly from the marathon data, and completed an individual written assessment covering all Sub-Strand 2.9 learning outcomes. |
| <b>How does this explain the phenomenon?</b> | Runner A won the 2023 Nairobi Marathon because of consistently high velocity maintained throughout the race, visible as a steep, nearly constant gradient on the displacement-time graph. Runner B's velocity decreased sharply between 25 km and 35 km — shown by a shallower gradient on the displacement-time graph and a negative slope (deceleration) on the velocity-time graph — creating a time deficit that Runner B's final sprint (positive acceleration in the last 2 km) could not overcome. This demonstrates that in linear motion, the shape of the graph — not just the start and end points — tells the full mathematical story of motion.                     |

| DIFFERENTIATION AND INCLUSION       |                                                                                       |                                                                                  |
|-------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Learner Need                        | Support Strategies                                                                    | Extension Strategies                                                             |
| English Language Learners           | Provide bilingual glossaries; allow use of first language for initial model drawing.  | Write extended explanations of observations; lead group discussions in English.  |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts.               |
| Gifted and advanced learners        | Access to additional reading materials; challenge questions during DQB.               | Lead model-building discussions; research and present extension topics to class. |